42028: Deep Learning and Convolutional Neural Network

Assignent-3 Project Report

Project Title: CNN-Based Multi-Domain Medical Image Screening for Skin, Eye, and Dental Disease Detection

Team Name: MediVision

Project Number: 85

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Abstract

This report presents MediVision, a deep-learning-based multi-domain medical image screening system that automatically classifies diseases across three clinical domains dermatology (skin), ophthalmology (eye), and dentistry (teeth) through a single web interface. The system implements a two-stage Convolutional Neural Network inference pipeline: Stage 1 is a domain classifier that detects whether an uploaded image is skin, eye, or dental, and Stage 2 routes the image to one of three specialised disease classifiers fine-tuned for that domain.

Two backbone architectures were rigorously evaluated VGG16 (the Custom CNN benchmark) and ResNet50 each instantiated as four separate models, yielding eight trained and benchmarked models in total. The unified training corpus comprises 35,884 deduplicated medical images spanning 24 disease classes, sourced from public Kaggle and Roboflow repositories under strict quality controls including PIL magic-byte verification and 16x16 perceptual-hash deduplication.

ResNet50 achieved the best end-to-end pipeline accuracy of 85.91% on a 5,648-image stress test, with per-domain accuracies of 83.94% (skin, 11 classes), 93.99% (eye, 6 classes), and 95.71% (teeth, 7 classes), and a domain routing accuracy of 99.45% on the full 5,648-image end-to-end stress test. VGG16 reached 84.03% end-to-end. ResNet50 was selected as the production backbone. The system is delivered as a Flask REST API plus a vanilla HTML / CSS / JavaScript single-page application featuring automatic domain detection, Grad-CAM heat-map explanations, calibrated abstention thresholds, three-tier severity triage (Urgent / Moderate / Mild), symptom-aware probability refinement, and a mandatory disclaimer-acknowledgement gate.

1. Introduction and Background

Medical imaging is central to the screening of a wide range of conditions: dermoscopic photographs identify malignant skin lesions, fundus and external-eye photography reveal cataracts and conjunctival inflammation, and dental photography exposes caries and gum disease. Access to qualified specialists, however, is uneven globally dermatologists are scarce outside major cities, ophthalmology wait-lists are long, and many populations face financial or geographic barriers to dental care. In parallel, Convolutional Neural Networks have matured to the point where they match or exceed expert performance on well-defined medical-imaging benchmarks. MediVision sits at this intersection: an accessible, browser-based, AI-assisted screening tool that serves as a first-line educational triage layer not replacing clinicians but helping users identify when professional evaluation is warranted.

Most existing CNN-based screening tools operate within a single domain (a skin classifier, or a retinopathy detector, or a caries detector). MediVision takes a different approach: a unified multi-domain pipeline in which the user simply uploads an image, and the system automatically detects whether it is skin, eye, or dental before routing it to the matching specialist classifier. This eliminates user-side decision-making and brings the workflow closer in spirit to how a general practitioner triages a presenting symptom.

1.1 The Problem to Be Solved

Three interlocking problems are addressed. First, the classification problem: can deep CNNs achieve credible accuracy on skin lesions, eye conditions, and dental anomalies using publicly available training data? Second, the routing problem: can a single CNN reliably distinguish between skin, eye, and dental images, given that the three domains differ dramatically in modality (close-up photography vs external-eye photography vs intraoral/X-ray)? Third, the safety problem: how can an AI screening tool be deployed responsibly to non-expert users when overconfident or unexplained predictions could cause real harm if patients act on them?

The technical answer to problems one and two is transfer learning ImageNet-pretrained CNNs learn low-level features that transfer well to medical imaging, even though ImageNet itself contains no medical images. The answer to problem three is a combination of three complementary mechanisms: Grad-CAM heat-map visualisation that exposes the spatial regions driving each prediction, calibrated abstention thresholds that flag low-confidence outputs for clinical follow-up, and a mandatory disclaimer-acknowledgement gate in the user interface that blocks all interaction on first visit.

1.2 Motivation

The motivation is grounded in both clinical need and educational value. Clinically, early detection of malignant skin lesions, eye disease, and dental caries dramatically improves outcomes and reduces treatment cost, yet the barrier to a specialist appointment often delays presentation until disease is advanced. A free, anonymous, browser-based screening tool that takes only an image and an optional symptom description could lower that barrier and prompt earlier consultation.

Educationally, the project allowed the team to engage end-to-end with the modern deep-learning workflow as practised in industry: data acquisition from heterogeneous public sources, rigorous deduplication and quality filtering, stratified splitting with leakage prevention, class-imbalance mitigation through focal loss and Neural Style Transfer augmentation, transfer learning with two-phase fine-tuning, cross-architecture benchmarking, calibrated abstention, Grad-CAM interpretability, and full-stack deployment as a Flask web application. Few undergraduate projects integrate this many production-grade techniques in a single coherent system.

1.3 Application

MediVision targets two user groups. The primary group is members of the public who notice a visible symptom a skin spot, an irritated eye, a discoloured tooth and want a quick, anonymous, educational opinion on whether to seek professional advice. They upload a photograph, optionally describe their symptoms in free text, and receive a ranked list of possible conditions, a confidence score with calibrated abstention, a Grad-CAM heat-map showing which region drove the prediction, a three-tier severity triage tag, and a plain-language explanation. The secondary group is students and researchers studying deep learning for medical imaging, for whom MediVision serves as a worked reference implementation of a complete multi-domain CNN pipeline.

The system is explicitly an educational and preliminary screening tool. A first-visit modal disclaimer blocks all content until acknowledged, every page carries an amber medical-advice banner, and the results screen prominently advises consultation with a qualified healthcare professional. The calibrated abstention threshold provides an additional safety layer by flagging low-confidence predictions with a red warning rather than presenting potentially misleading certainty.

1.4 Dataset

The unified MediVision corpus was assembled from seven distinct public sources across Kaggle and Roboflow, merged into a single training pipeline with strict quality controls. After deduplication the corpus contains 35,884 medical images spanning 24 disease classes across three medical domains, with an additional 4,500-image stratified domain training subset drawn from the disease pools.

The dataset composition evolved substantially from the initial proposal (Part A) and implementation plan (Part B) as the project matured. The Part B plan described a unified corpus of approximately 43,000 images spanning 17 disease classes (8 skin, 4 eye, 5 teeth) drawn primarily from ISIC 2019, ODIR-5K, and two dental radiography sets, with a single multi-task CNN architecture using shared domain and disease output heads. Several changes were made during implementation in response to empirical findings and data-quality analysis. First, the class taxonomy was expanded from 17 to 24 disease classes: the skin domain grew from 8 to 11 classes by incorporating additional dermatological conditions from the Skin Diseases Image Dataset (Kaggle); the eye domain expanded from 4 to 6 classes by adding Dry Eye (sourced via Roboflow object-detection crops), Eyelid Drooping, and Uveitis from the Mendeley Eye Diseases Classification dataset; and the teeth domain grew from 5 to 7 classes by merging three Kaggle oral-disease datasets and supplementing the Normal class from a Roboflow teeth-domain dataset. Second, the total corpus size decreased from approximately 43,000 to 35,884 images despite the addition of new sources, because the perceptual-hash deduplication step (described below) removed 5,855 duplicate images that were inflating the original count — the Part B estimate of 43,000 included these duplicates. Third, the single multi-task CNN architecture proposed in Part B was replaced with a two-stage pipeline of four independent models (one domain router plus three disease classifiers), which allowed per-domain optimisation of input resolution, augmentation policy, and loss function without cross-domain interference. Fourth, the architecture comparison shifted from Custom CNN / ResNet50 / InceptionV3 (as tested in the Part C initial experiments, where baseline accuracies ranged from 43–81%) to VGG16 / ResNet50 for the final system, because VGG16 provided a stronger "Custom CNN" benchmark than the from-scratch baseline used in Part C, and InceptionV3 was dropped in favour of the more widely adopted ResNet50 which outperformed it in our domain after proper fine-tuning with the expanded dataset and improved training strategy.

Quality assurance was implemented in two stages. First, PIL magic-byte format detection forced a full pixel decode on every image, rejecting files with misleading extensions (e.g. WebP saved as .jpg) and catching truncated files. Second, a 16x16 grayscale average-perceptual-hash deduplication step removed both exact duplicates and trivial re-encodings that would have inflated validation metrics. Deduplication removed 976 skin duplicates, 27 eye duplicates, and 4,852 teeth duplicates particularly important for the teeth corpus where the source datasets had significant overlap.

MediVision - Dataset Summary

Task	Source	Classes	Train	Val	Test	Total
Domain	sampled from disease p	3	3150	675	675	4500
Skin Disease	ismailpromus/skin-disea	11	23906	4321	4322	32549
Eye Disease	Mendeley (5 classes) +	6	1940	416	416	2772
Teeth Disease	rajapriyanshu/teeth-dat	7	4244	909	910	6063

Table 1. *Unified dataset composition post-deduplication: 35,884 disease images across 24 classes plus a 4,500-image domain-routing subset stratified-sampled from the disease pools.*

A 70 / 15 / 15 train / validation / test split was applied across the unified corpus, stratified by class label and verified to be free of path-level leakage between splits. For the skin domain the dominant Melanocytic Nevus class was capped at 3,500 training samples to prevent gradient dominance, and seven minority skin classes were upsampled to 2,000 training samples each. For the eye domain the underrepresented Dry Eye and Uveitis classes were augmented with 150 synthetic images each, generated by VGG19-based Neural Style Transfer that blended disease-relevant content structure with style textures sampled from within-class exemplars. Class-weight compensation (sklearn balanced) was applied to all classifiers.

Figure 1. *Representative samples from each medical domain (top: skin; middle: eye; bottom: teeth). Within-domain visual heterogeneity motivates per-domain augmentation policies and per-domain input resolutions.*

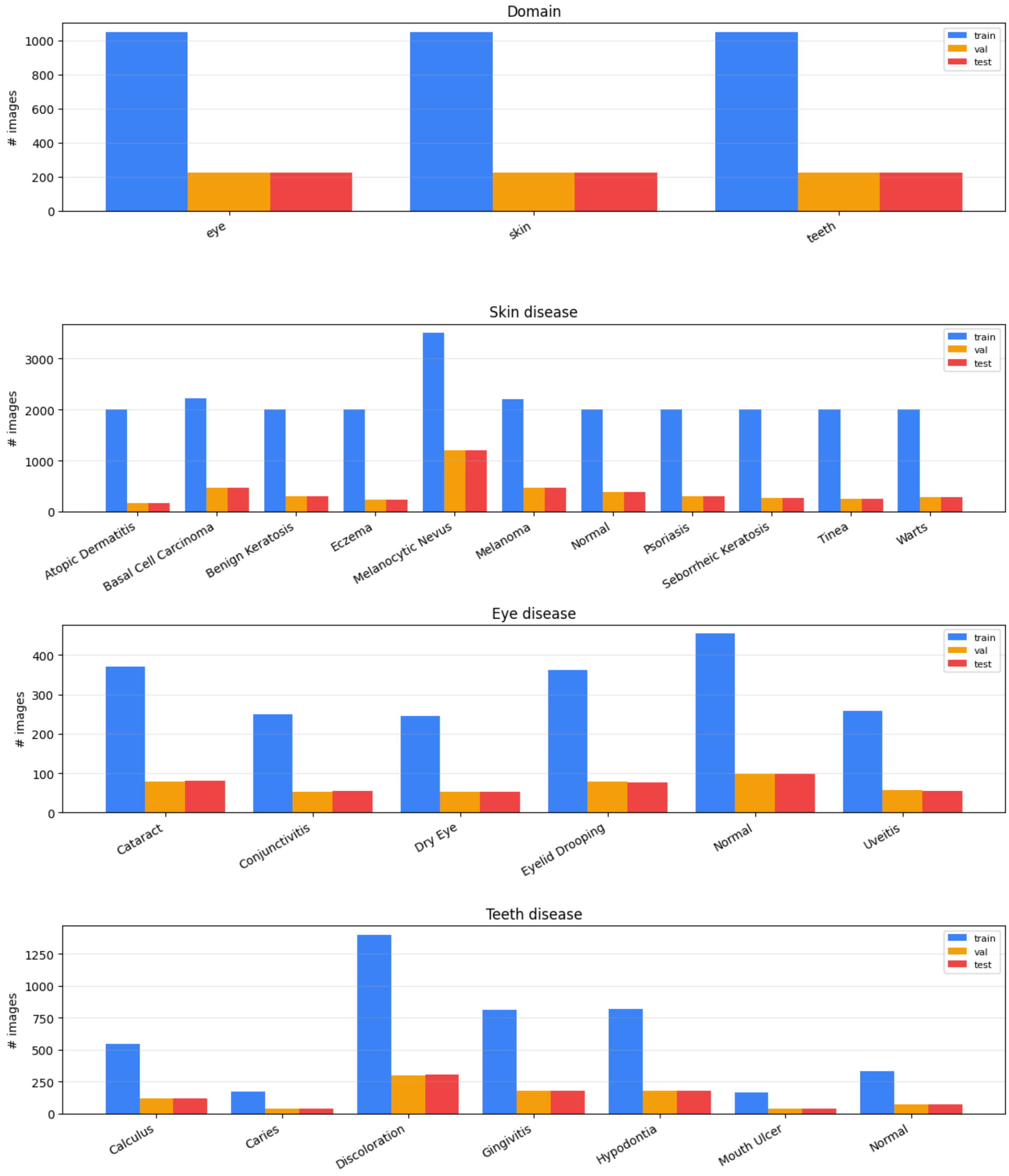

Figure 2. *Class distribution across train (blue) / validation (orange) / test (red) splits for all four classification tasks. The skin domain shows substantial pre-rebalancing imbalance; the post-rebalancing training counts shown reflect the capping and upsampling described in Section 1.4.*

2. Overview of the Architecture / System

MediVision is built around a two-stage neural inference pipeline that combines a domain router with three domain-specific disease classifiers. This design was chosen over a single 24-class flat classifier for three reasons. First, each disease classifier can be optimised independently with domain-specific preprocessing, augmentation, and input resolution (the eye classifier uses 384x384 inputs; the others use 224x224). Second, different loss functions per domain focal loss for the severely imbalanced skin domain, categorical cross-entropy elsewhere do not compromise the others. Third, the system is modular: individual disease classifiers can be retrained or swapped without disturbing the others, an important operational consideration.

2.1 System Workflow

The pipeline runs as follows. The user uploads a medical image through the web interface. The image is forwarded to the Stage 1 domain classifier, which produces a probability distribution over the three domain labels (skin / eye / teeth) and routes the image via argmax. The Stage 2 disease classifier for the predicted domain then returns a softmax distribution over that domains conditions. If the user has supplied free-text symptoms, a post-hoc symptom-refinement module applies weighted keyword matching with a capped probability boost, allowing the system to favour conditions whose symptom signatures match the description without overriding visual evidence. Grad-CAM is then computed on the disease model, and the final response carries the ranked predictions, the Grad-CAM heat-map, a severity triage tier, and a plain-language explanation.

The pipeline is implemented as the following ordered stages:

1. Image upload user supplies a JPG, PNG, or WEBP image (max 10 MB). Input validation rejects non-image MIME types and oversize files before they reach the model.
2. Preprocessing the image is resized to 224x224 (or 384x384 for the eye disease classifier), converted from RGB to BGR with ImageNet mean subtraction, and assembled into a batch tensor of shape (1, H, W, 3).
3. Stage 1 domain classification: the preprocessed image is passed through the domain classifier, which emits a 3-element softmax distribution.
4. Routing argmax over the Stage 1 output selects the appropriate Stage 2 disease classifier and re-preprocesses the image at the disease classifiers native resolution if different.
5. Stage 2 disease classification: the disease classifier emits a softmax over that domains classes. Test-time augmentation (TTA) is optionally applied by averaging predictions over the original image, a horizontal flip, and three randomly shifted crops (five forward passes total).
6. Calibrated abstention If the top probability falls below the calibrated threshold for that domain (0.80 skin, 0.60 eye, 0.80 teeth), the system flags the prediction as low-confidence rather than displaying a confident-looking result.
7. Symptom refinement (optional) the symptom module performs case-insensitive weighted keyword matching against per-disease symptom lists. Matched diseases receive a probability boost capped at +0.25 to prevent text from overriding visual evidence.
8. Output rendering the GUI displays the predicted condition with confidence, the top-k ranked predictions, the Grad-CAM overlay, the severity triage tier, and a templated plain-language explanation.

To verify that the routing layer generalises beyond the stratified sample it was trained on, the domain classifier was stress-tested on the concatenated disease test sets of all three domains a total of 5,648 images. The classifier achieved 99.45% routing accuracy in this stress test, confirming that the high training accuracy is not an artefact of stratified sampling.

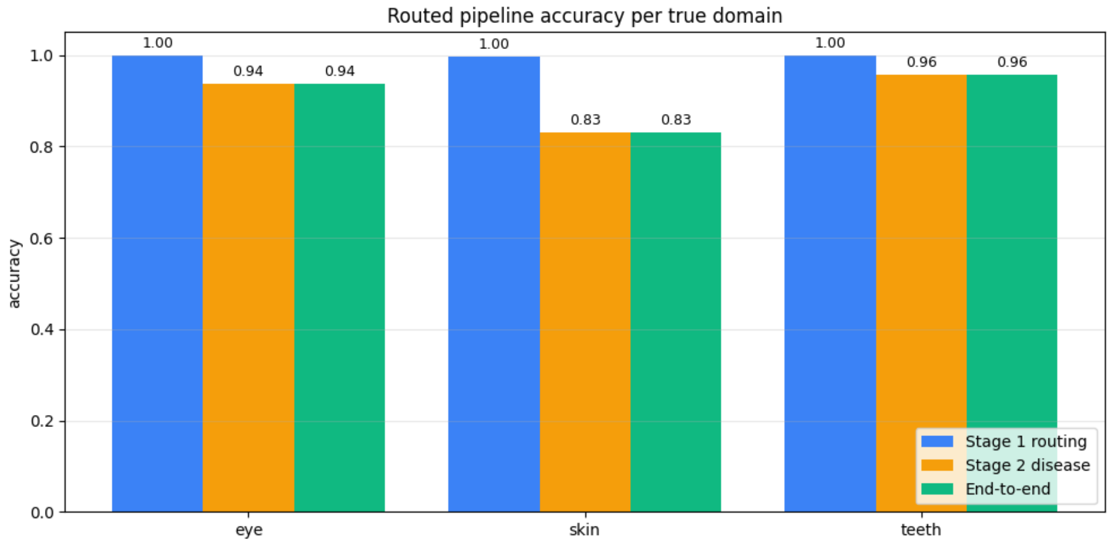

Figure 3. *Routed pipeline accuracy per true domain on the 5,648-image end-to-end stress test. Stage 1 routing maintains >99% accuracy across all three domains; end-to-end accuracy is rate-limited by Stage 2 (lowest on skin at 0.83).*

2.2 CNN Architecture Design

Two backbone architectures were instantiated for each of the four models (domain router plus three disease classifiers), producing eight trained models in total for cross-architecture comparison. Both backbones were pre-trained on ImageNet and fine-tuned on the medical data using single-phase or two-phase training depending on dataset size and class-imbalance characteristics.

VGG16 (Custom CNN) Backbone

VGG16, introduced by Simonyan and Zisserman (2014), is a 16-layer CNN organised into five convolutional blocks, each comprising two or three 3x3 Conv2D layers with ReLU activation followed by 2x2 MaxPooling. Channel depth grows from 64 in the first block to 512 in the last, producing a final 7x7x512 feature map from a 224x224x3 input. The ImageNet classification head was discarded and replaced with Global Average Pooling, Dropout 0.4, a Dense(256) layer with ReLU activation and L2 weight regularisation of 1e-4, a second Dropout 0.4, and a final Dense softmax layer sized to the target classes. Trainable parameters range from approximately 14.85M (3-class domain) to 14.85M (11-class skin).

ResNet50 Backbone

ResNet50 (He et al., 2016) is a 50-layer residual network organised into four stages of bottleneck blocks. Each block applies a 1x1 convolution to reduce channel depth, a 3x3 convolution at the reduced depth, a 1x1 convolution to restore depth, and a residual skip connection that adds the block input back to its output. The skip connections allow gradient signals to flow directly from later layers to earlier ones, dramatically alleviating the vanishing-gradient problem that limits trainable depth in plain CNNs like VGG16. The network produces a 7x7x2048 feature map from a 224x224x3 input four times deeper in channels than VGG16. The custom head (GAP + Dropout

0.4 + Dense 256 ReLU + Dropout 0.4 + Dense softmax) is identical to that used with VGG16 for direct comparison; total trainable parameters are approximately 23.6M.

Why ResNet50 Was Selected for Production

Although both backbones achieved comparable per-domain accuracies, ResNet50 was selected for the deployed web application. The decision rests on five independent considerations summarised in Table 2.

Criterion	VGG16 (Custom CNN)	ResNet50 (Production)
End-to-end pipeline accuracy	84.03%	**85.91% (+1.88 pp)**
Skin classification (hardest)	82.18% (Macro F1 0.753)	**83.94% (Macro F1 0.775)**
Domain routing accuracy	98.96%	**99.56% (+0.60 pp)**
Trainable parameters	~14.85M	~23.6M
Architecture depth	16 layers (plain stack)	50 layers (residual)
Gradient flow	Plain forward propagation	Skip connections
Feature-map channel depth	7x7x512	7x7x2048 (4x richer)
Calibration (teeth abstain rate)	15.60%	**11.65%**

Table 2. *ResNet50 versus VGG16 - production-suitability comparison. ResNet50 wins on accuracy, calibration, and feature richness; VGG16 has a smaller parameter footprint.*

Superior pipeline accuracy. ResNet50 outperforms VGG16 on the end-to-end metric by 1.88 pp (85.91% versus 84.03%). On the hardest task (11-class skin) ResNet50 wins on raw accuracy (83.94% versus 82.18%) and on macro F1 (0.775 versus 0.753). The macro F1 gain is especially important because it indicates better performance on the minority malignant classes (Melanoma, BCC) that the screening tool most needs to surface.

Better residual gradient flow. VGG16s plain stack suffers from vanishing gradients during full fine-tuning, especially on small datasets like the eye domain. ResNet50s skip connections produce smoother training curves and more reliable convergence.

Richer feature representation. ResNet50s final feature map is four times deeper in channels than VGG16, providing more fine-grained texture information for distinguishing visually similar conditions (e.g. Benign Keratosis versus Seborrheic Keratosis).

Better confidence calibration. On the teeth domain ResNet50 abstains on 11.6% of predictions versus 15.6% for VGG16 meaning ResNet50 is confident on a larger fraction of test images while maintaining comparable accuracy on the confident subset.

Industry-standard backbone. ResNet50 is the de facto baseline in published medical-imaging work. Standardising on it makes the deployment easier to compare against external benchmarks and easier to maintain.

Training Strategy

- Single-phase training (Domain, Skin, Teeth): the full ImageNet-pretrained backbone was unfrozen from epoch 0 and trained end-to-end at LR 1e-4. Suitable because these datasets were large enough (>4,500 training samples) to support full fine-tuning without overfitting. Up to 100 epochs with patience 20.

- Two-phase transfer learning (Eye): because the eye dataset was the smallest (1,940 training samples after augmentation), Phase 1 froze the backbone and trained only the classification head for 15 epochs at LR 1e-3; Phase 2 unfroze the top 50 layers and continued for up to 85 epochs at LR 1e-5, avoiding catastrophic forgetting of pre-trained representations.
- Common configuration: Adam optimiser, label smoothing 0.1, 3-epoch linear LR warm-up, L2 1e-4 on the Dense head, Dropout 0.4, sklearn balanced class weights, ReduceLROnPlateau on val loss.
- Focal loss for skin: the 11-class skin classifier used focal loss (gamma=2.0, alpha=0.25) instead of categorical cross-entropy, down-weighting easy-to-classify examples to focus capacity on the rare malignant classes.

Parameter	Domain	Skin	Eye	Teeth
Backbone (production)	ResNet50	ResNet50	ResNet50	ResNet50
Input resolution	224x224x3	224x224x3	384x384x3	224x224x3
Output classes	3	11	6	7
Training strategy	Single-phase	Single-phase	Two-phase TL	Single-phase
Loss function	Label-smoothed CE	Focal (g=2.0)	Label-smoothed CE	Label-smoothed CE
Max epochs / patience	100 / 10	100 / 20	15 + 85 / 20	100 / 20
Learning rate	1e-4	1e-4	1e-3 -> 1e-5	1e-4
Optimiser	Adam	Adam	Adam	Adam
Dropout / L2	0.4 / 1e-4	0.4 / 1e-4	0.4 / 1e-4	0.4 / 1e-4
Batch size	32	32	16	32
Label smoothing	0.1	0.1	0.1	0.1
Augmentation policy	Generic	Off (upsampling)	Strong (eye-spec.)	Generic

Table 3. *Per-model architecture and training configuration. Eye uses 384x384 inputs and two-phase TL; skin uses focal loss and minority upsampling instead of online augmentation.*

2.3 GUI Design

The MediVision web application is delivered as a Flask REST API backend with a vanilla HTML / CSS / JavaScript frontend no client-side framework, no build step. This deliberately minimal stack keeps the deployment footprint small, maximises cross-browser compatibility, and avoids the operational overhead of a JavaScript build pipeline. The backend loads all four ResNet50 checkpoints once at server startup and keeps them resident across requests, achieving sub-second inference latency. Uploaded images are processed in memory only via io.BytesIO and are never written to disk, supporting a strong privacy posture appropriate for a medical screening context.

The interface uses a clinically appropriate palette of deep teal (#1A4A5E) and medical green (#7ECFB3), with amber caution banners for medical disclaimers. Times New Roman is used for headings to convey formal medical documentation aesthetics, while a system sans-serif is used for body copy to maintain on-screen readability. Pictographic emoji were replaced with inline SVG line icons during the polish pass, producing a cleaner clinical aesthetic. The application comprises three core screens (Landing, Upload, Results) plus a How It Works section and an About section accessible from the navigation bar.

Screen 1: Landing Page

The landing page is the entry point and consent gate. A first-visit modal disclaimer blocks all interaction until the user clicks I Acknowledge; consent is then stored in sessionStorage so the modal only appears once per session. After acknowledgement the user sees a hero section with the system title and a Begin Screening CTA, three domain capability cards (Skin / Eye / Dental), an amber disclaimer banner, and a secondary I Understand & Proceed button. Each domain card now lists the number of conditions detected and the most clinically important example classes (e.g. Skin: 11 conditions including Melanoma, Basal Cell Carcinoma, Eczema, Psoriasis, and Tinea), with a Selected automatically footer reinforcing that there is no manual choice to make.

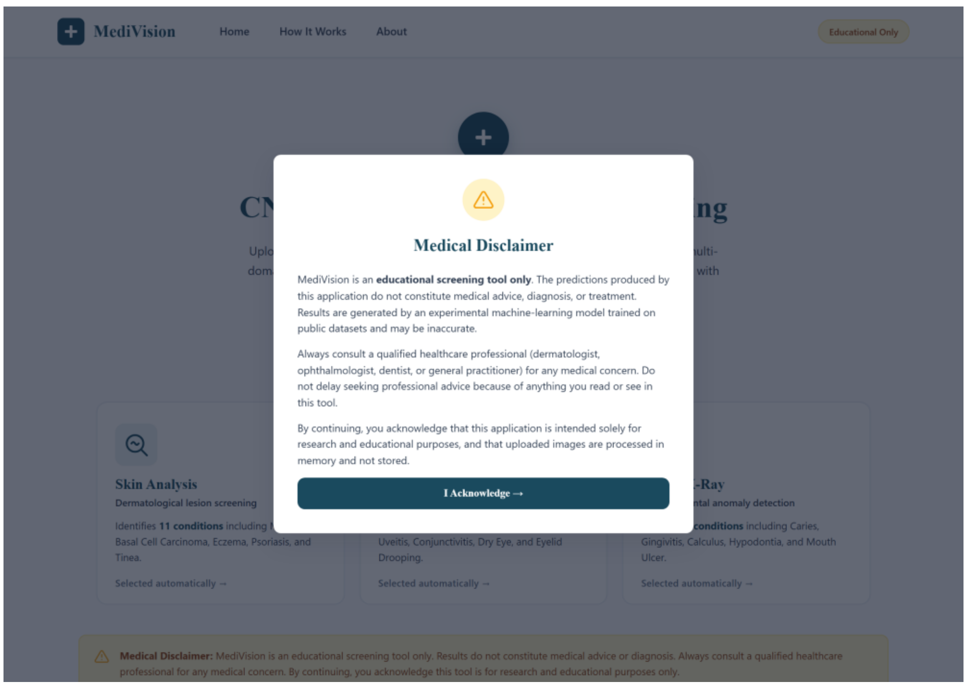

Figure 4. *First-visit medical disclaimer modal. All interaction is blocked behind a semi-transparent overlay until acknowledgement; consent is stored in sessionStorage.*

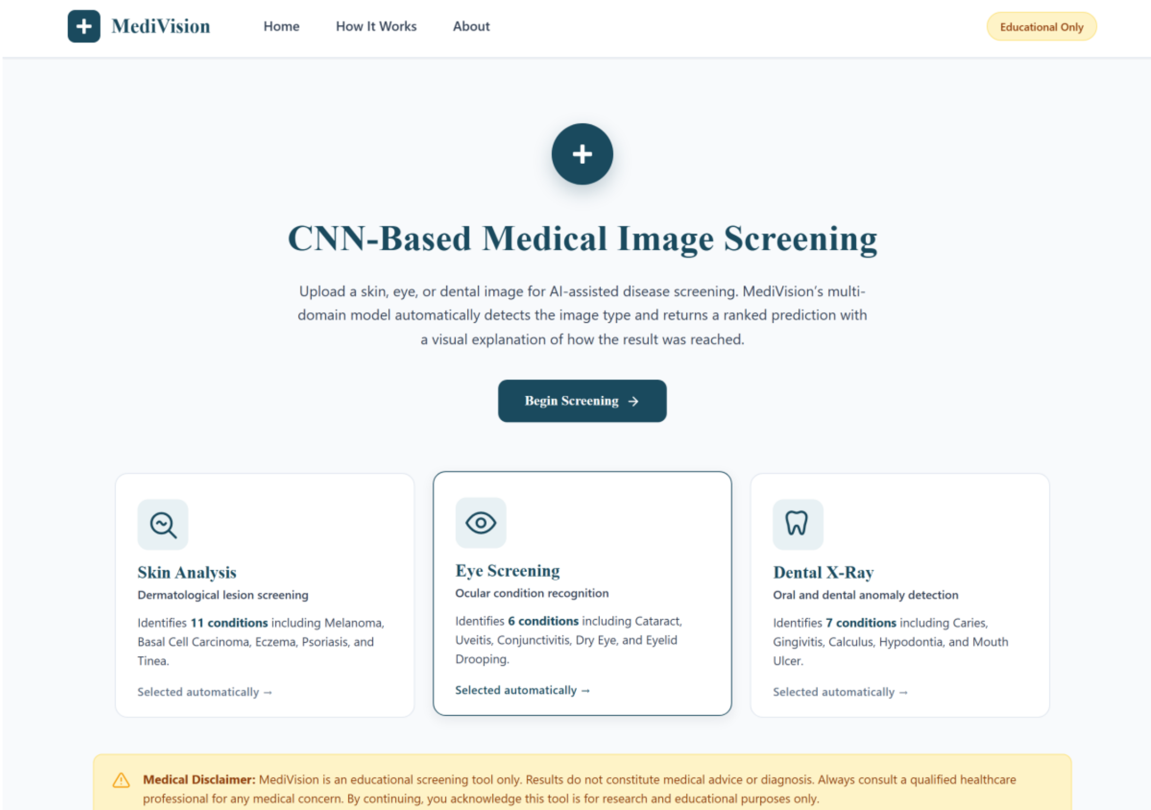

Figure 5. *Landing page hero with the SVG medical-cross brand mark, three domain capability cards, and the amber disclaimer banner. Pictographic emoji from earlier drafts were replaced with inline SVG line icons during the design polish.*

Screen 2: Image Upload Screen

The upload screen is the primary data-entry interface. A blue Auto-detection enabled info banner reassures the user that no manual domain selection is required. The drag-and-drop zone accepts JPG, PNG, or WEBP up to 10 MB; client-side validation immediately rejects non-image MIME types or oversize files with a red toast. Once a file is selected a thumbnail preview card displays the filename, file size, and a green Loaded badge. An optional symptoms textarea lets the user describe their observation in free text. The primary Analyze Image button is disabled until a valid file is loaded; on click a loading-spinner overlay appears while the API request is in flight.

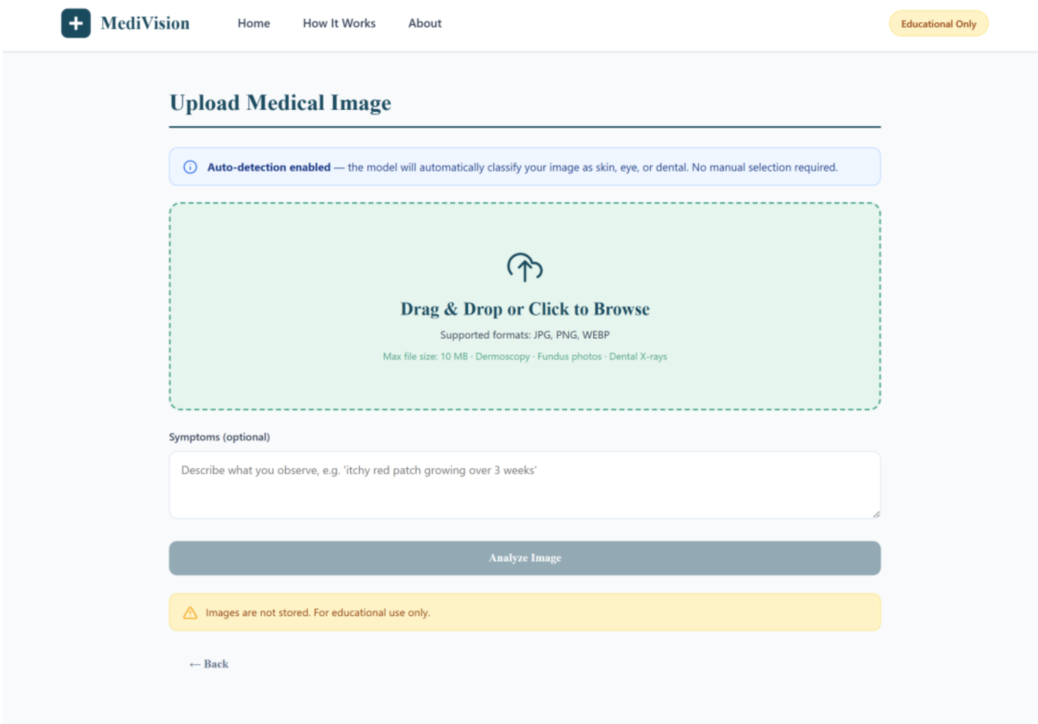

Figure 6. *Upload screen in its empty state: auto-detection info banner, dashed drag-and-drop zone with cloud-upload SVG icon, optional symptoms textarea, and the disabled Analyze Image CTA.*

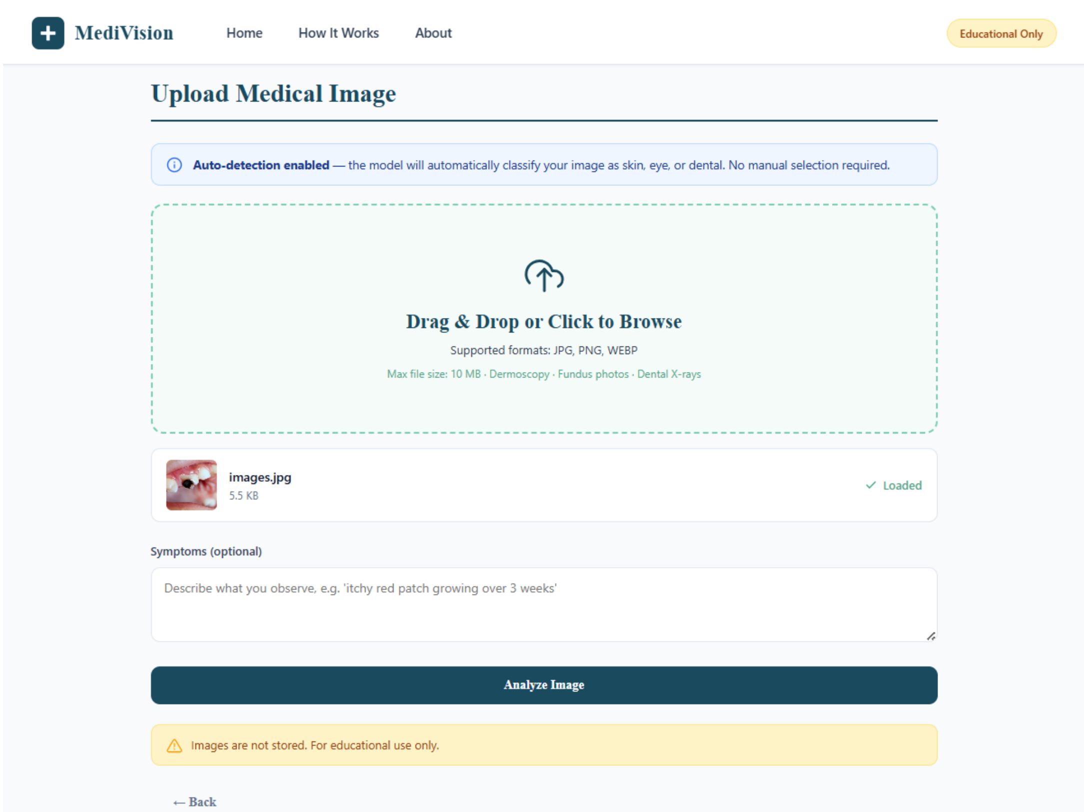

Figure 7. *Upload screen after a valid file selection. The preview card appears with thumbnail, filename, file size, and the green Loaded badge; the symptoms textarea has been used, and Analyze Image is now enabled.*

Screen 3: Results and Explanation Screen

The results screen is the most information-dense view. The header shows two coloured pills: a green Domain badge (e.g. Skin Domain) summarising the Stage 1 routing decision, and a blue Confidence badge (e.g. 87.4% Confidence) summarising the models certainty. If the confidence falls below the calibrated abstention threshold the confidence pill turns red and a Low Confidence Warning banner is rendered above the metric cards. Two metric cards show the Predicted Condition with a one-line lay description, and the Confidence Score with a teal progress bar.

Below the metrics, the Top Predictions horizontal bar chart visualises every class in the predicted domain, with the top prediction in teal, URGENT-severity classes in red, and others in soft teal-grey. To its right, the Grad-CAM Heatmap panel displays the base-64 PNG returned by the API, accompanied by a red-to-blue activation legend and an explanatory caption. If symptoms were provided, a Symptom Refinement panel lists which user phrases matched which classes. The severity banner is colour-coded by tier (red Urgent, amber Moderate, green Mild / Normal) and carries a one-sentence recommendation. A green Plain-Language Explanation box generates a 2-3 sentence summary client-side from a template using the predicted condition, confidence, and domain. Two action buttons close out the screen: Download Report (browser-print API with a print-specific CSS) and Analyze New Image (resets state, returns to Upload).

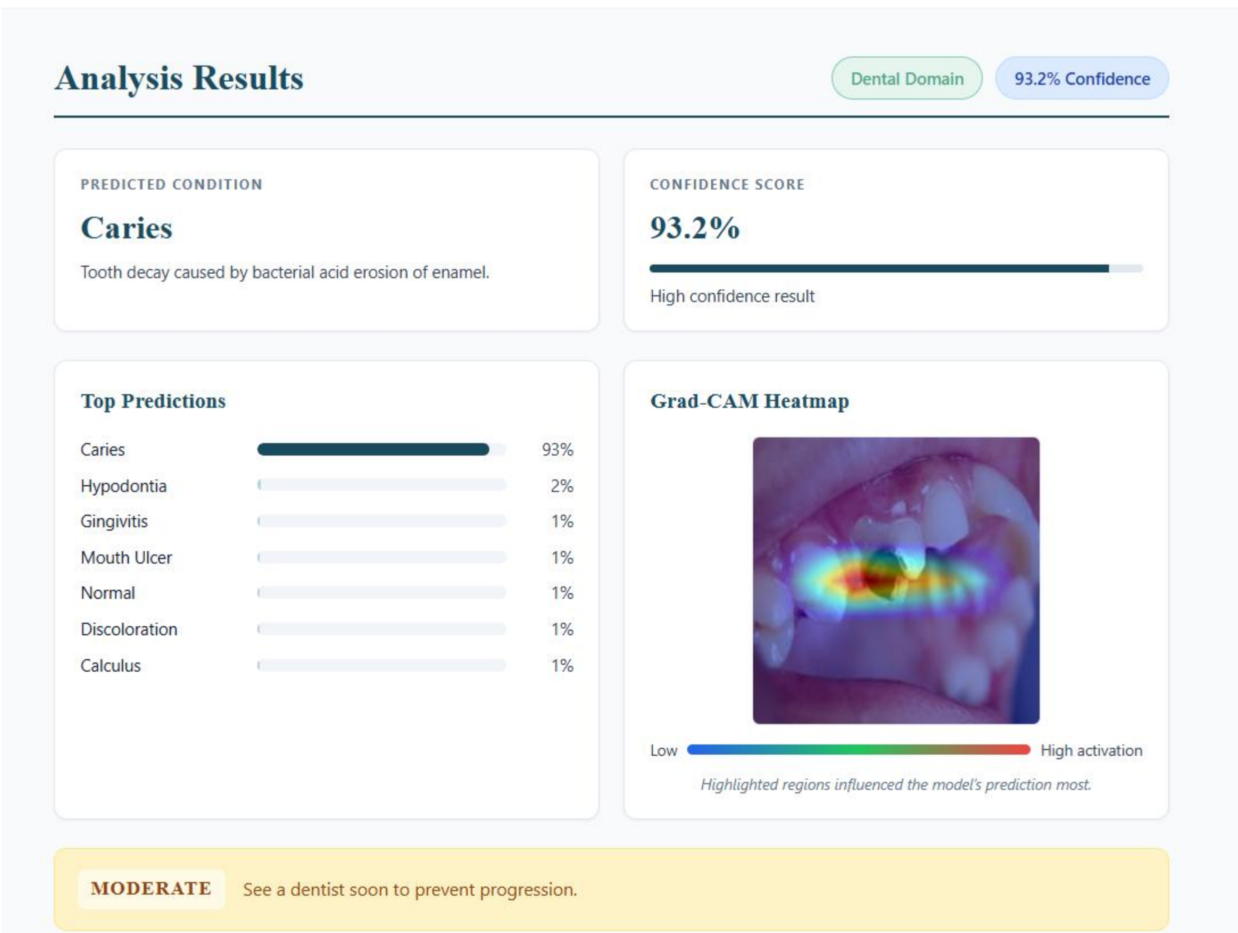

Figure 8. *Results screen top fold: Analysis Results header with domain and confidence pills, two metric cards, Top Predictions horizontal bar chart, and the Grad-CAM panel with red-to-blue activation legend.*

How It Works Section

The How It Works section, anchored from the navigation, explains the four-step pipeline in user-friendly language: (1) Upload Image, (2) Automatic Routing via the ResNet50 domain classifier, (3) Disease Classification via the domain-specific fine-tuned ResNet50, and (4) Visual Explanation via Grad-CAM heat-map plus severity triage. Each step is presented in a numbered card with an SVG icon, a heading, and a one-sentence description, demystifying the pipeline for non-technical users.

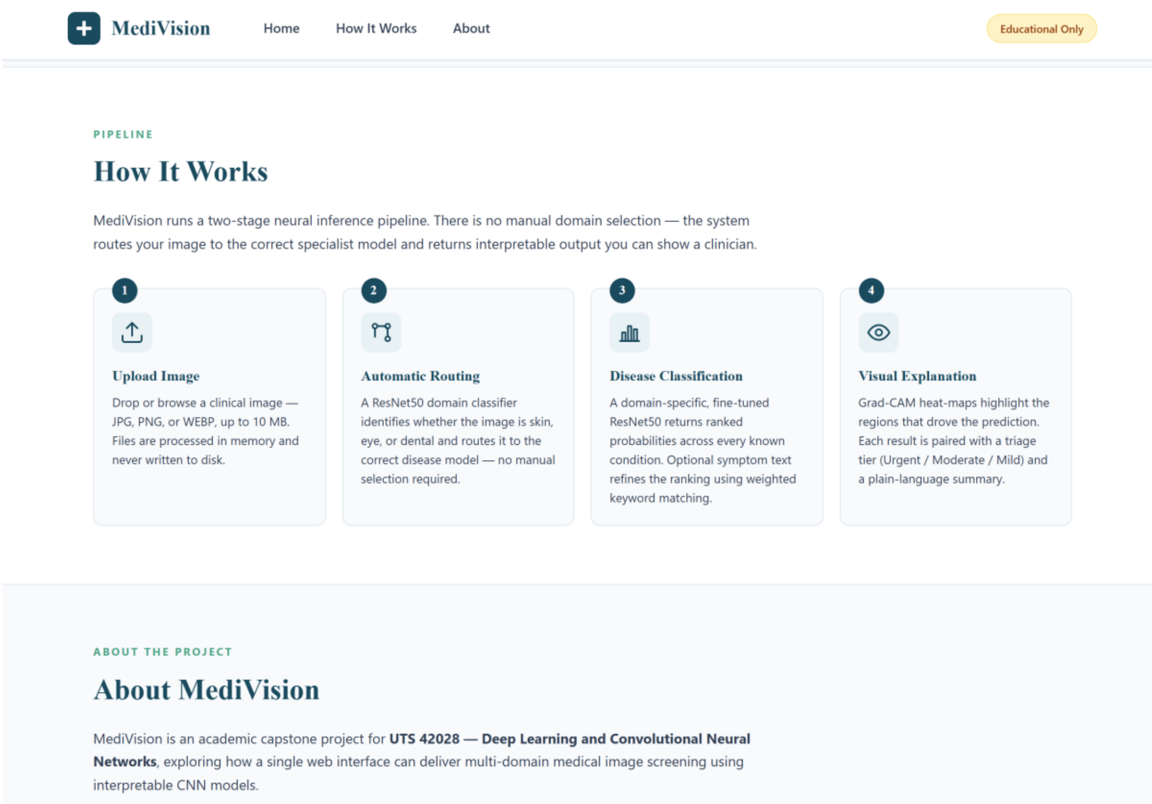

Figure 9. *How It Works section: four numbered step cards (Upload Image -> Automatic Routing -> Disease Classification -> Visual Explanation) with monochrome SVG line icons.*

About Section

The About section identifies the system as an academic capstone for UTS 42028 - Deep Learning and Convolutional Neural Networks, summarises the architecture and interpretability stack across four mini-cards (Project, Architecture, Interpretability, Datasets), and lists the team members with their student IDs. A research-only footer reinforces that MediVision is not a registered medical device.

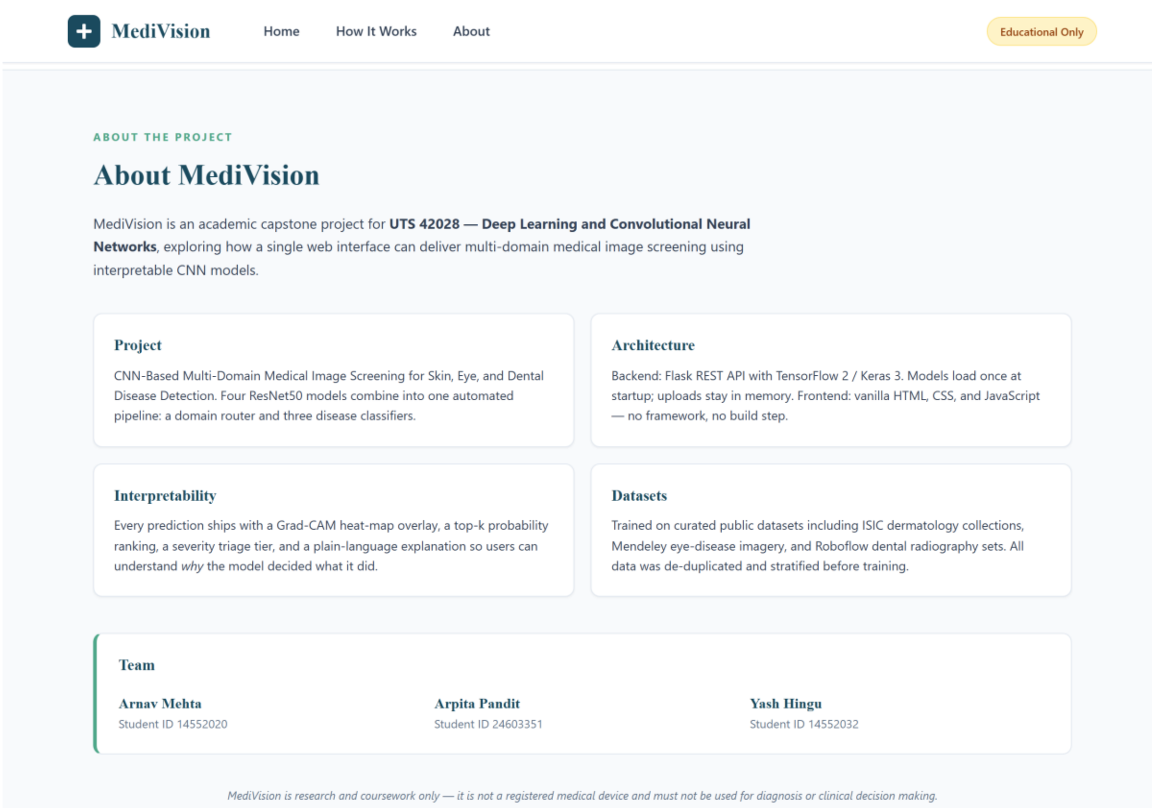

Figure 10. *About section: project context, four-card architecture summary, team member block, and a research-only footnote stating that MediVision is not a registered medical device.*

3. Results and Evaluation

3.1 Experimental Settings

All experiments were conducted on Google Colab Pro with an NVIDIA A100 GPU (40 GB VRAM). The software stack comprised TensorFlow 2.20.0, Keras 3, NumPy 2.0.2, pandas 2.2.2, scikit-learn for class-weight computation and stratified splitting, OpenCV for Grad-CAM heat-map rendering, and Pillow for image verification. TF32 mixed precision was implicitly enabled on the A100 Tensor Cores, accelerating matrix multiplications by approximately 8x relative to FP32.

Reproducibility and recoverability were treated as first-class concerns. A custom checkpoint system implemented atomic weight writes (write-to-temp + rename), mirrored every checkpoint to Google Drive, and used an architecture fingerprint (num_classes + class_names + img_size + model_tag) to refuse to silently load incompatible weights. A 0-byte weight file was treated as missing rather than as a corrupted checkpoint. The training loop emitted a verbose Resume Report before every fit() call declaring the current epoch, best validation accuracy, and which weight files were loaded. This system survived multiple Colab session disconnects and instance preemptions without losing any training state.

Hyper-parameters were defined in a single configuration cell at the top of each notebook for clarity and reproducibility: 100 epochs maximum, patience-based early stopping (10 for domain, 20 for disease classifiers), label smoothing 0.1, a 3-epoch linear LR warm-up, ReduceLROnPlateau, and Top-K = 3 for top-k accuracy reporting. Evaluation produced bootstrap 95% confidence intervals on test accuracy by resampling the test set 1,000 times with replacement, providing a principled estimate of result variability.

3.2 Pre-Processing

- Image verification PIL in strict mode opens every file, checks the magic byte, and forces a full pixel decode, rejecting truncated or mislabelled files. This step alone rejected two WebP-as-JPG files in the skin corpus.
- Perceptual-hash deduplication each image is reduced to a 16x16 grayscale thumbnail, the mean intensity computed, and each pixel binarised against the mean to produce a 256-bit perceptual hash. Identical hashes group together; only the first image in each group is retained. This caught both exact duplicates and trivial re-encodings, removing 976 skin, 27 eye, and 4,852 teeth duplicates (a 28% reduction on the teeth corpus that materially improved evaluation honesty).
- Stratified splitting with leakage prevention each cleaned dataset was split 70 / 15 / 15 train / val / test, stratified by class label, then sanity-checked via assertions: (a) no filepath appears in more than one split; (b) every class has at least five samples in val and test; (c) label encoders are fully populated with no NaN. Any failure aborted the notebook before GPU time was wasted.
- Resizing and channel normalisation all images were resized to 224x224 (or 384x384 for the eye disease classifier) and passed through the ResNet50 / VGG16 preprocess_input transform, which converts RGB to BGR and subtracts ImageNet channel means [103.939, 116.779, 123.68].

- Online augmentation applied inside the tf.data pipeline using tf.keras.layers preprocessing layers, in the [0, 255] colour space before preprocess_input (otherwise RandomContrast and RandomBrightness behave incorrectly on the BGR-shifted tensors). A generic policy (h-flip, +/-10% rotation, +/-10% zoom, +/-10% contrast, +/-5% translation) was used for domain / skin / teeth; a stronger eye-specific policy added brightness and contrast jitter. Augmentation was selectively disabled for skin because minority upsampling already provided regularisation.
- Class-imbalance mitigation three complementary strategies: (1) the dominant Melanocytic Nevus class was capped at 3,500 training samples; (2) minority skin classes were upsampled to 2,000 each via sklearn-style duplication; (3) sklearn compute_class_weight balanced weights were passed to model.fit() for every classifier.
- Neural Style Transfer augmentation (eye) for the two most underrepresented eye classes (Dry Eye and Uveitis, with about 50 and 75 images before augmentation) a VGG19-based NST pipeline generated 150 synthetic images per class. Each synthetic image preserved disease-relevant content while adopting the style of another within-class exemplar.

3.3 Experimental Results

This section presents the headline cross-architecture comparison, followed by per-domain detail (training curves, confusion matrices, ROC curves) and the Grad-CAM interpretability analysis.

Model / Task	Test Acc.	Top-3 Acc.	Macro F1	Wtd. F1	Macro AUC	Abstain
VGG16 Domain	98.96%	100.00%	0.990	0.990	-	2.2%
VGG16 Skin	82.18%	96.23%	0.753	0.823	0.976	16.4%
VGG16 Eye	93.75%	99.52%	0.925	0.937	0.996	8.4%
VGG16 Teeth	94.95%	99.12%	0.909	0.947	0.988	15.6%
VGG16 Pipeline (E2E)	84.03%	-	-	-	-	-
ResNet50 Domain	**99.56%**	100.00%	0.996	0.996		1.0%
ResNet50 Skin	**83.94%**	96.74%	0.775	0.841	0.981	17.2%
ResNet50 Eye	**93.99%**	99.76%	0.928	0.939	0.996	13.5%
ResNet50 Teeth	**95.71%**	99.45%	0.925	0.955	0.986	11.6%
ResNet50 Pipeline (E2E)	**85.91%**	-	-	-	-	-

Table 4. *Final test results - VGG16 versus ResNet50 across all five tasks. ResNet50 wins every comparison, with the largest gains on the hardest task (11-class skin) and the end-to-end pipeline.*

Domain Classifier Detail

The domain classifier is the foundation of the pipeline if it routes incorrectly, the disease classifier downstream cannot recover. ResNet50 achieved 99.56% test accuracy on the held-out 675-image domain test set (95% bootstrap CI [98.96%, 99.93%]). The confusion matrix shows only three misclassifications, all involving skin at the boundary with eye/teeth consistent with the observation that some close-up clinical skin photographs resemble dermatoscopic eye-area imagery.

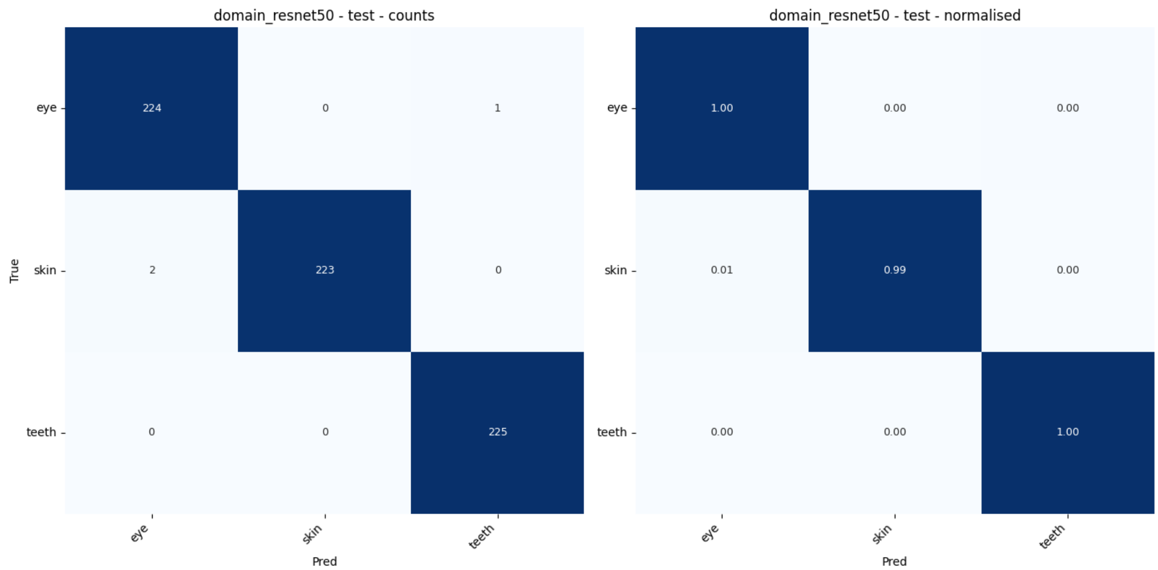

Figure 11. *ResNet50 domain-classifier confusion matrix (counts left, normalised right). Only 3 of 675 test images are misclassified, all at the skin boundary.*

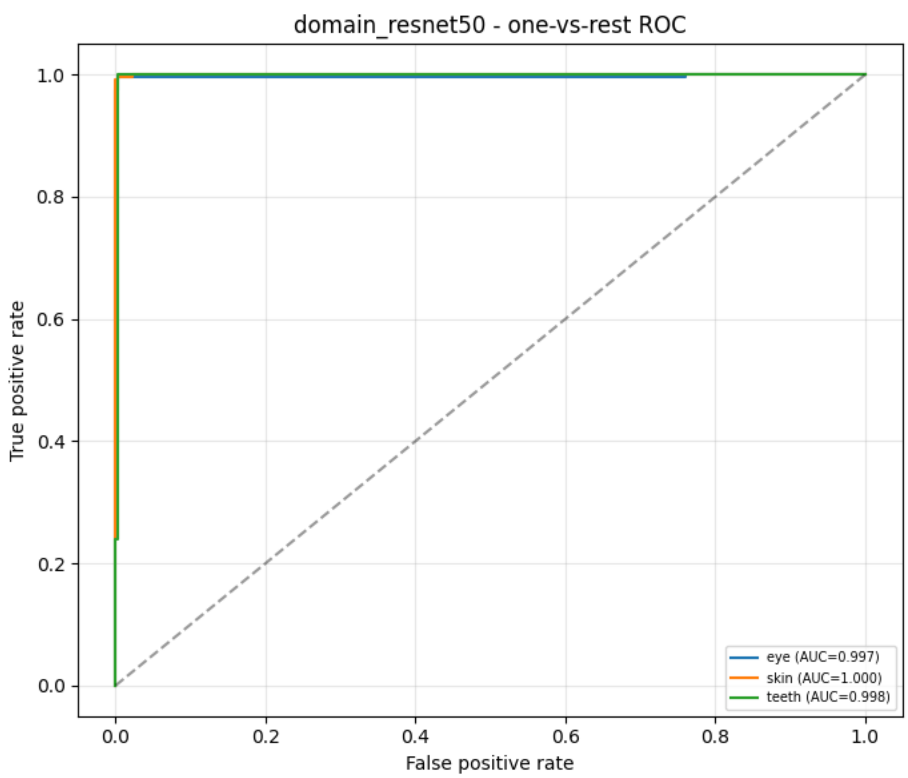

Figure 12. *ResNet50 domain-classifier one-vs-rest ROC. Macro AUC is effectively 1.0; all three classes achieve AUC >= 0.997.*

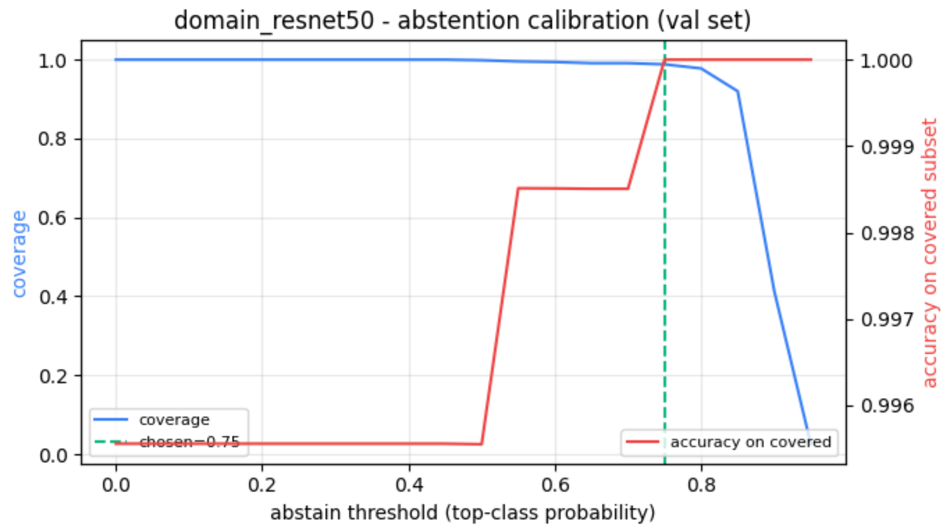

Figure 13. *Abstention-calibration curve for the ResNet50 domain classifier. Coverage (blue, left axis) and accuracy on the covered subset (red, right axis) trace out the operating curve; the green dashed line marks the chosen threshold (0.75), at which abstention is minimal but covered accuracy is near-perfect.*

Skin Disease Classifier Detail

The 11-class skin classifier is the hardest task due to high inter-class visual similarity and a long-tail class distribution. ResNet50 achieved 83.94% test accuracy (95% CI [82.79%, 85.12%]) with top-3 accuracy of 96.74% meaning the correct class is in the top three predictions 97 times out of 100. This top-3 metric is particularly relevant for a screening tool, where presenting the top three differential diagnoses is more clinically useful than insisting on a single prediction. Per-class performance is excellent for the malignant classes (Melanoma F1 0.97, BCC F1 0.92) and the well-represented Melanocytic Nevus (F1 0.95), but weaker on the visually similar inflammatory dermatoses (Atopic Dermatitis F1 0.56, Psoriasis F1 0.62).

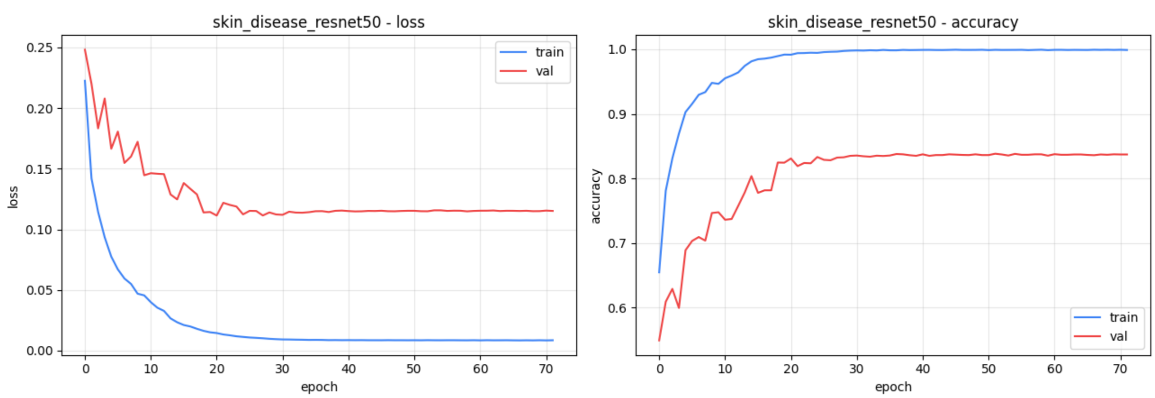

Figure 14. *Training curves for the ResNet50 skin disease classifier - loss (left) and accuracy (right). Train and val curves track closely; best validation accuracy of 84.0% is reached around epoch 27 with no significant overfitting.*

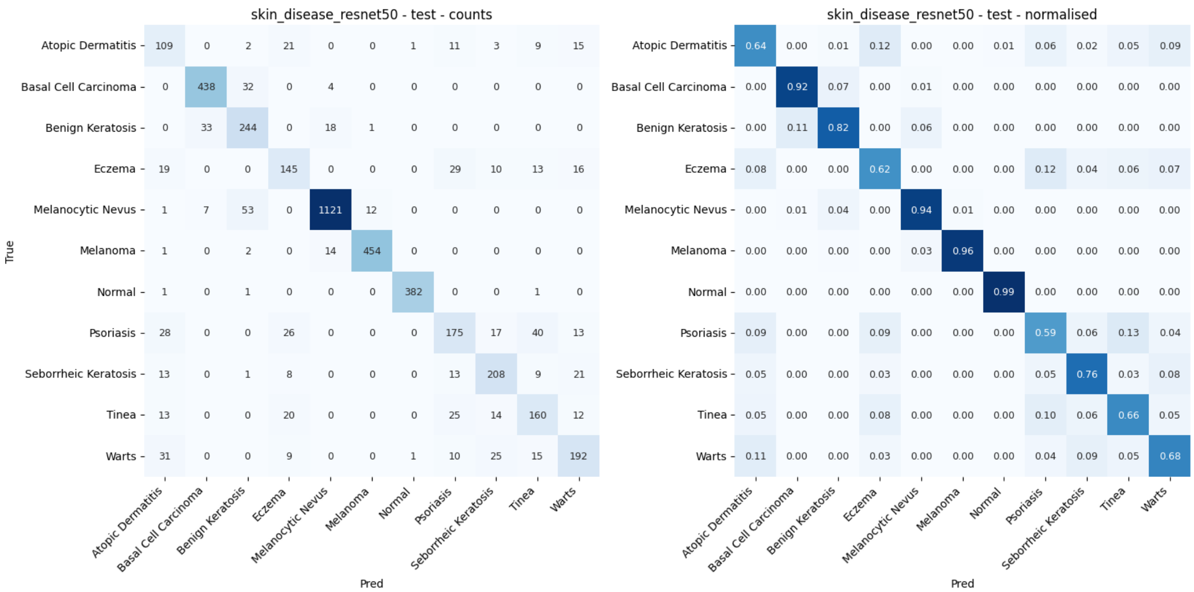

Figure 15. *ResNet50 skin disease confusion matrix (counts left, normalised right). Diagonal dominance is clear for malignant classes (Melanoma 0.96, BCC 0.92) and Normal (0.99); the largest off-diagonal mass is the Atopic Dermatitis / Eczema confusion (both inflammatory dermatoses with overlapping presentation).*

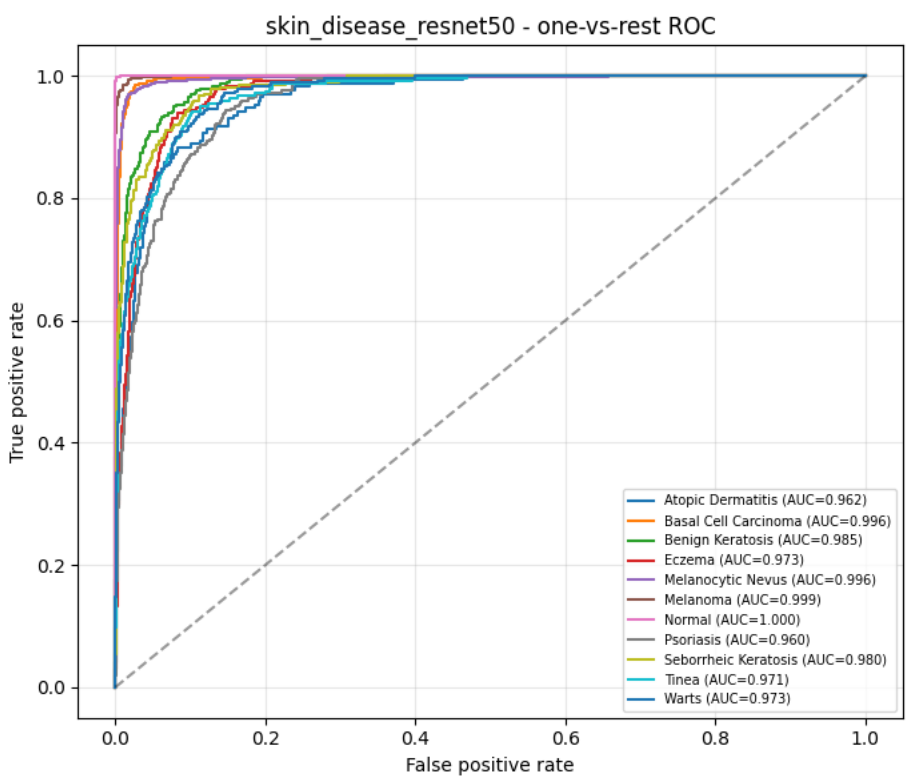

Figure 16. *Per-class one-vs-rest ROC for the ResNet50 skin disease classifier. Macro AUC = 0.981; every individual class achieves AUC > 0.96, with Melanoma at 0.999 and Normal at 1.000.*

Eye Disease Classifier Detail

The 6-class eye classifier achieved 93.99% test accuracy (95% CI [91.59%, 96.39%]) with top-3 accuracy 99.76%. This was the smallest dataset (1,940 training samples) and required the most aggressive interventions: 384x384 input resolution, two-phase transfer learning (frozen Phase 1 then 50-layer unfreeze in Phase 2), Neural Style Transfer augmentation for the Dry Eye and Uveitis minority classes, and a stronger eye-specific augmentation policy. The result is a classifier with uniformly strong per-class performance every class achieves F1 >= 0.86. The only systematic confusion is between Conjunctivitis and Uveitis, two inflammatory eye conditions with visually similar redness patterns.

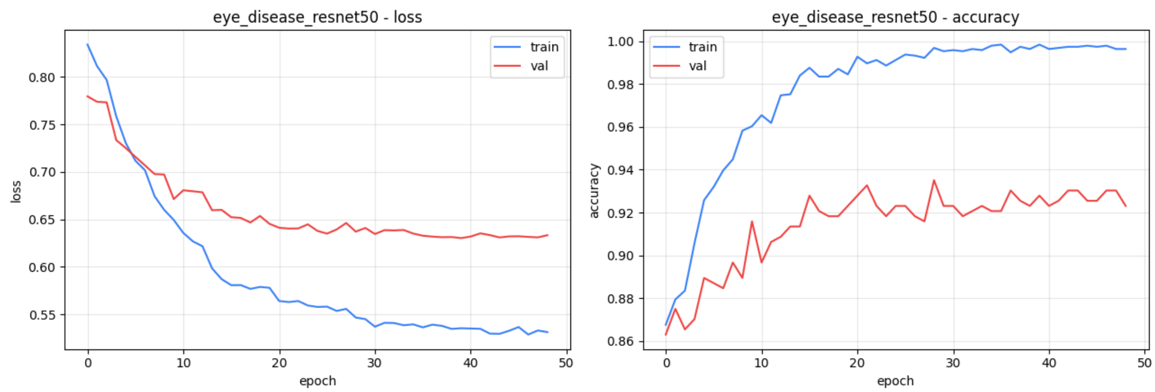

Figure 17. *Training curves for the ResNet50 eye disease classifier (two-phase TL). The vertical break around epoch 15 marks the Phase 1 -> Phase 2 transition; Phase 2 lifts val accuracy from ~89% to ~94%.*

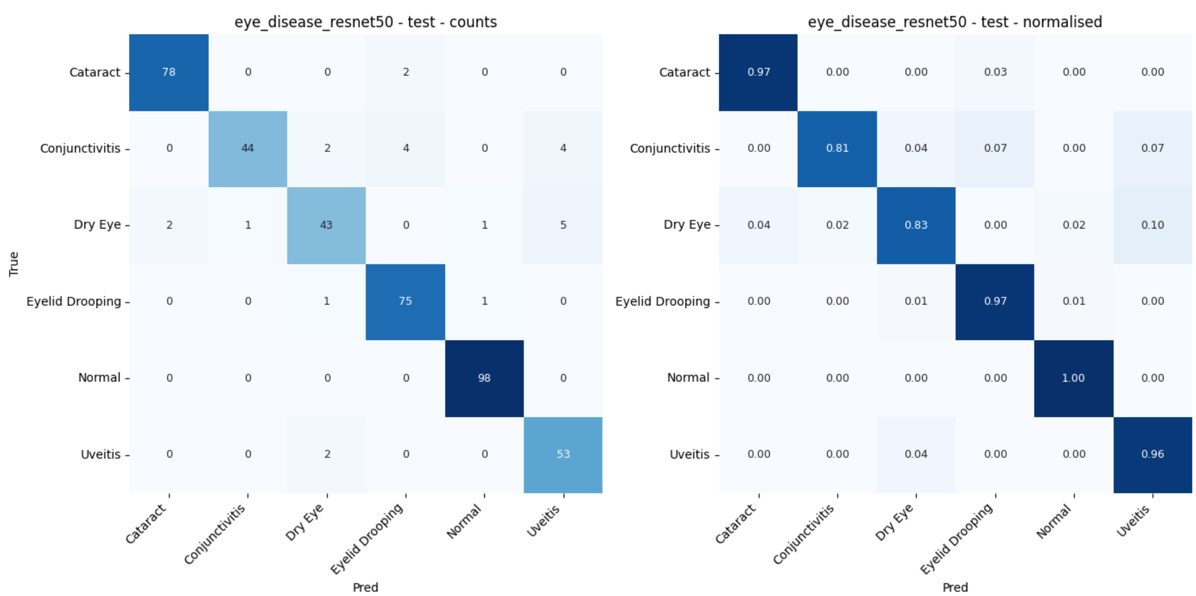

Figure 18. *ResNet50 eye disease confusion matrix (counts left, normalised right). All six classes show strong diagonals; the only notable confusion is Conjunctivitis <-> Uveitis (about 8% off-diagonal each direction).*

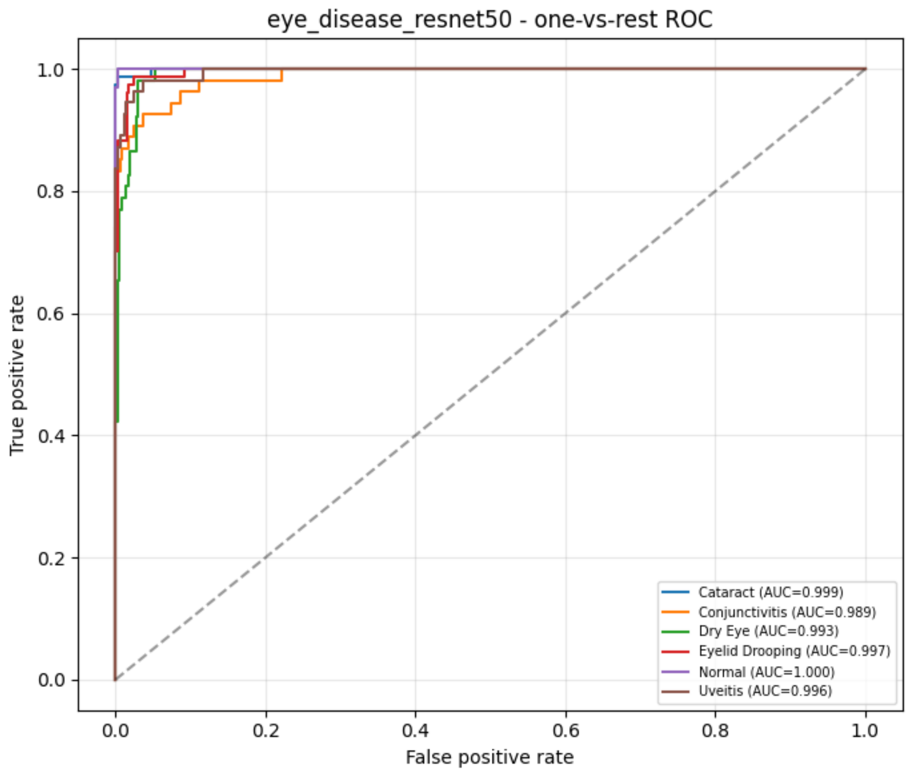

Figure 19. *Per-class one-vs-rest ROC for the ResNet50 eye disease classifier. Macro AUC = 0.996, with Normal and Cataract achieving near-perfect AUC = 1.000.*

Teeth Disease Classifier Detail

The 7-class teeth classifier achieved 95.71% test accuracy (95% CI [94.29%, 96.93%]) the highest of the three disease classifiers. Class-weight compensation and minority-class supplementation from a Roboflow Normal source produced excellent results for the well-represented classes (Discoloration F1 0.99, Hypodontia F1 0.98, Gingivitis F1 0.93, Calculus F1 0.91, Normal F1 0.99, Mouth Ulcer F1 0.99). The single weakness is the Caries class, which has only 36 test samples and a recall of 0.53 the model achieves perfect precision on Caries (1.00) but misses about half of true Caries cases. This is a known clinical limitation that future work should address by sourcing additional Caries training data.

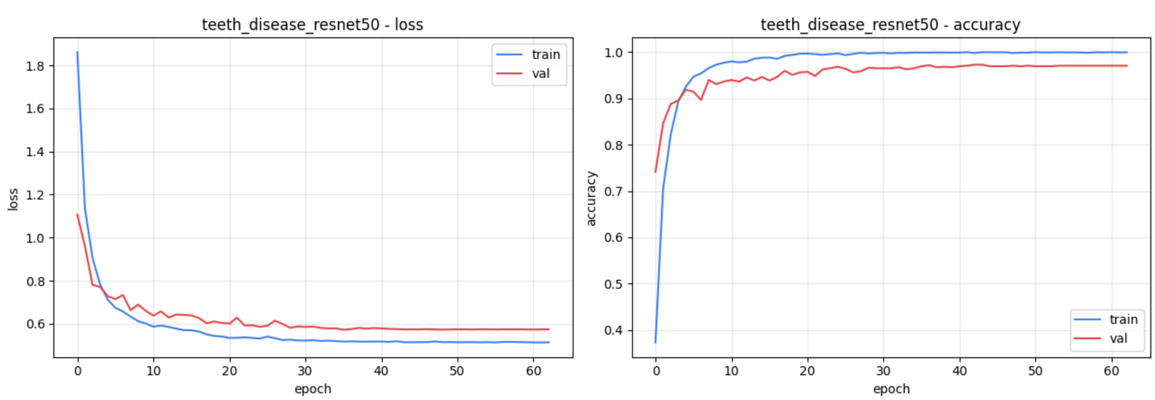

Figure 20. *Training curves for the ResNet50 teeth disease classifier. Best val accuracy of 95.93% reached around epoch 39 with stable convergence.*

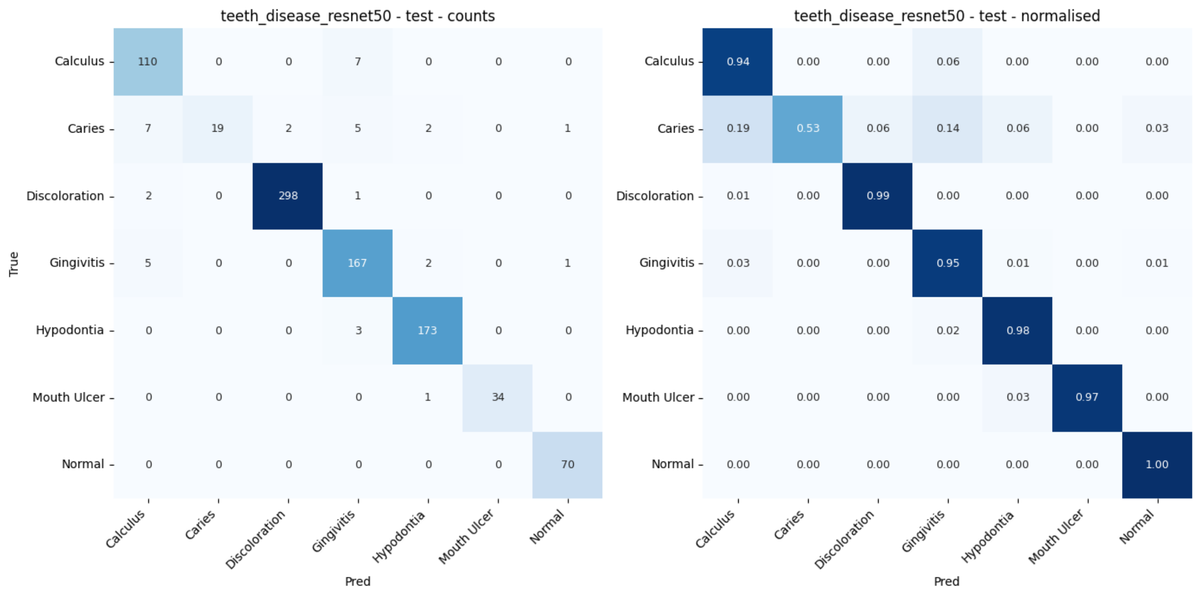

Figure 21. *ResNet50 teeth disease confusion matrix (counts left, normalised right). The Caries class carries the most off-diagonal mass; some Caries cases are misclassified as Calculus or Gingivitis (visually similar dental anomalies).*

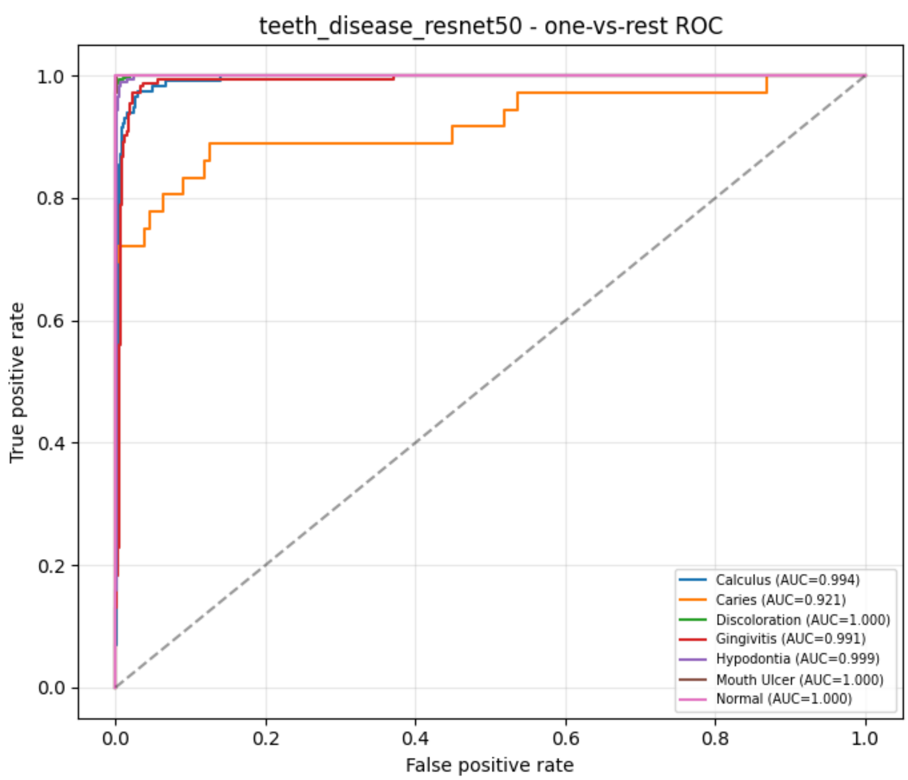

Figure 22. *Per-class one-vs-rest ROC for the ResNet50 teeth disease classifier. Macro AUC = 0.986, with all classes achieving AUC >= 0.96.*

Grad-CAM Interpretability Analysis

Grad-CAM (Gradient-weighted Class Activation Mapping) heat-maps were generated for correctly classified test images to verify that the models attend to clinically meaningful regions rather than spurious background features. The technique computes gradients of the predicted class score with respect to the final convolutional feature maps and weights the channel activations by their pooled gradient, producing a localised saliency map that is upsampled and overlaid on the input image. The skin classifier consistently attends to lesion boundaries (Figure 23a), the eye classifier attends to the iris and pupil region (Figure 23b), and the teeth classifier attends to the enamel surface and gum line (Figure 23c), confirming that each model has learned to localise the diagnostic feature rather than exploit dataset-level artefacts.

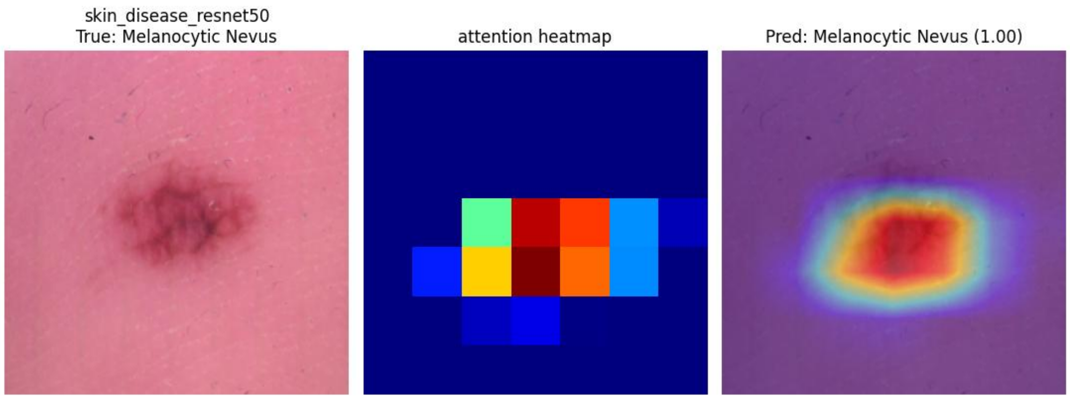

Figure 23a. *Grad-CAM on the ResNet50 skin classifier (Melanocytic Nevus, predicted 1.00). The attention concentrates on the pigmented lesion centre the diagnostically informative region.*

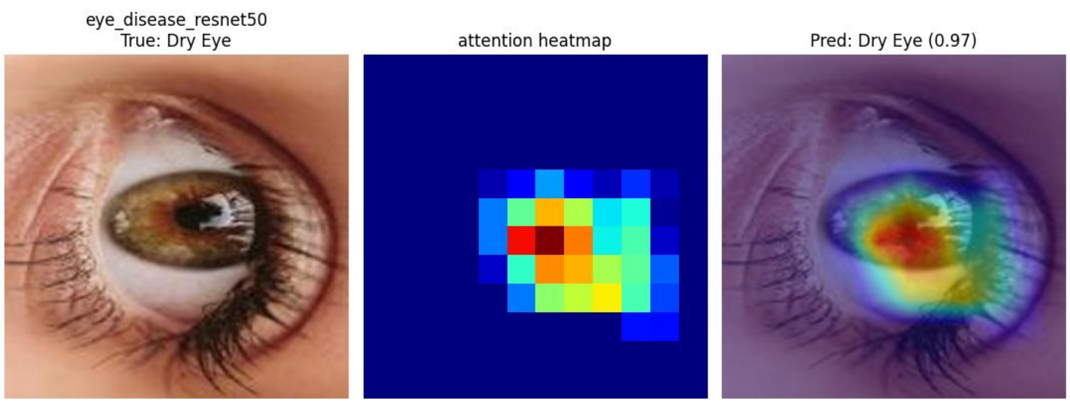

Figure 23b. *Grad-CAM on the ResNet50 eye classifier (Dry Eye, predicted 0.97). The activation concentrates on the iris / inner canthus region where dry-eye signs manifest clinically.*

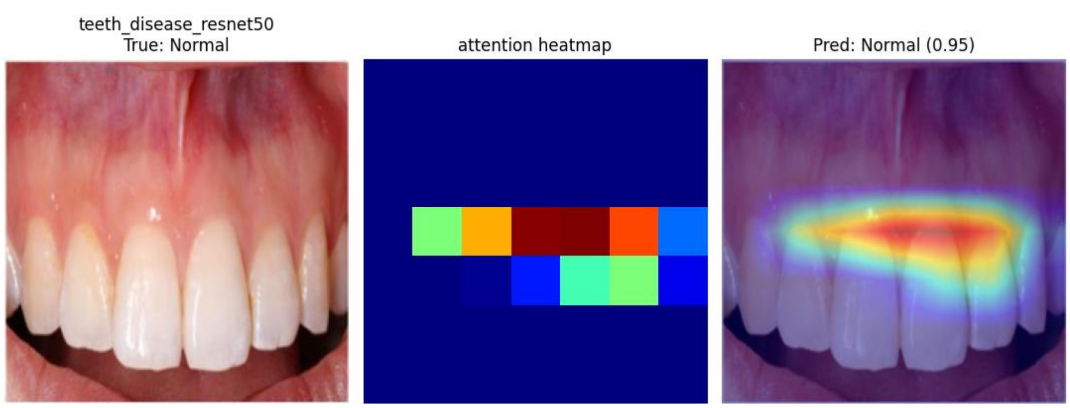

Figure 23c. Grad-CAM on the ResNet50 teeth classifier (Normal, predicted 0.95). Attention concentrates on the gum line and tooth row, the regions most informative for a Normal classification.

Calibrated Abstention Mechanism

A calibrated abstention mechanism flags low-confidence predictions for follow-up rather than risking misleading the user. For each disease classifier, a domain-specific confidence threshold was selected on the validation set such that abstaining from below-threshold predictions would maximise accuracy on the remaining confident subset while abstaining on no more than about 20% of predictions. The calibrated thresholds are 0.80 (skin), 0.60 (eye), and 0.80 (teeth). At these thresholds ResNet50 abstains on 17.2% of skin predictions but reaches 91.4% accuracy on the confident subset; abstains on 13.5% of eye predictions with 98.3% accuracy on the confident subset; and abstains on 11.6% of teeth predictions with 98.1% accuracy on the confident subset. The abstention status is surfaced directly in the GUI: when a prediction falls below the threshold the confidence pill turns red and the system displays a warning, rather than presenting a confident-looking prediction the user might act on inappropriately.

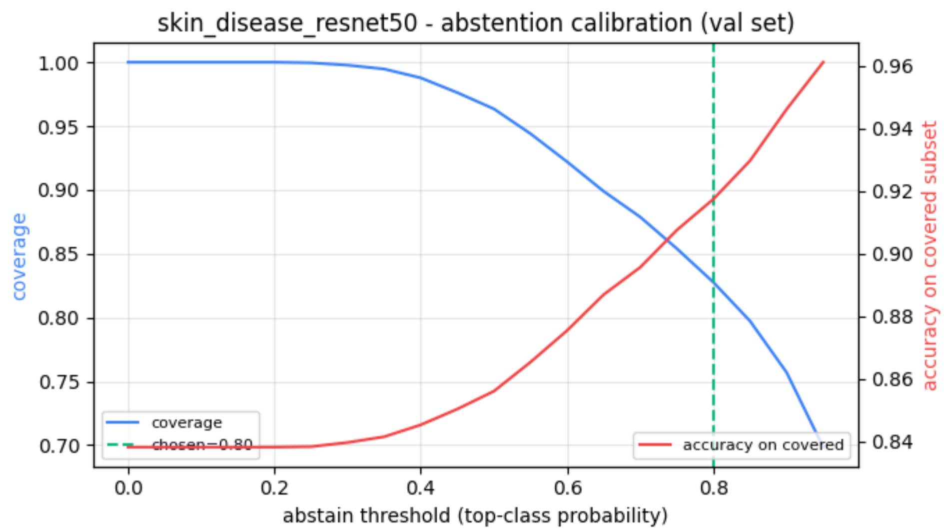

Figure 24. Abstention-calibration curve for the ResNet50 skin classifier. Coverage drops as the threshold rises (blue, left axis) while accuracy on the covered subset rises (red, right axis). The chosen threshold (green dashed line, 0.80) sits at the inflection point where covered accuracy crosses ~91%.

End-to-End Pipeline Evaluation

To evaluate the integrated system, the three disease test sets were concatenated into a unified end-to-end test set of 5,648 images, and each image was passed through the full Stage 1 -> Stage 2 pipeline. Stage 1 routing accuracy was 99.45% (ResNet50; only 31 of 5,648 images routed to the wrong disease classifier), and Stage 2 disease accuracy on the routed images was 85.91%. Because Stage 1 is so close to 100%, the end-to-end accuracy is effectively the average disease accuracy weighted by domain test-set size.

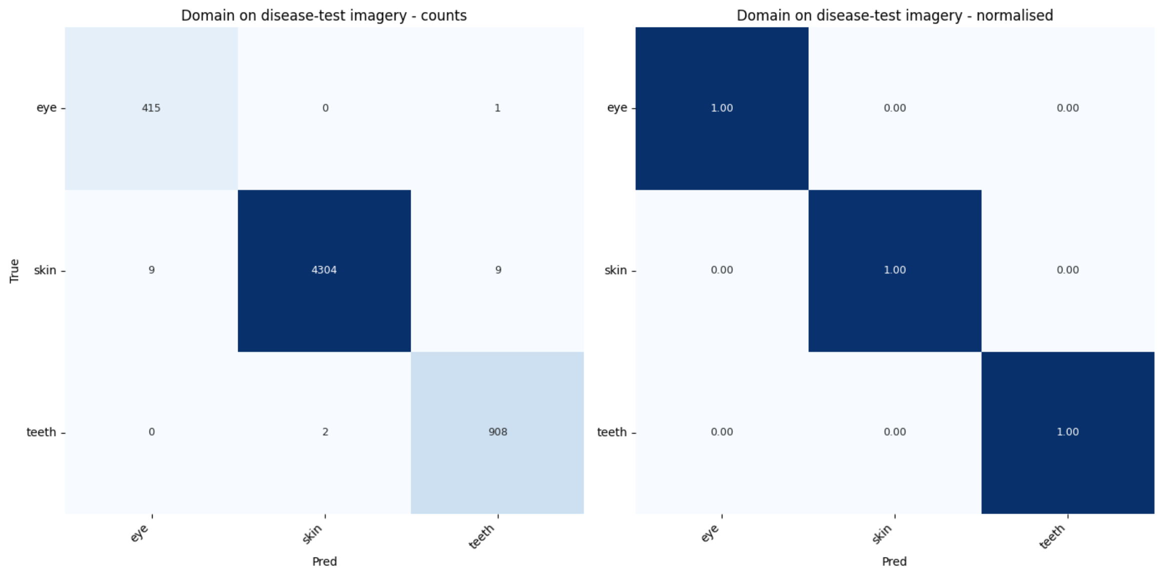

Figure 25. Domain confusion matrix on the concatenated disease test sets (5,648-image end-to-end stress test). All 31 routing errors involve the skin domain at its boundary with eye/teeth.

3.4 Limitations

Several limitations were identified through systematic evaluation. They are presented here transparently because honest assessment of failure modes is essential for any medical AI system.

- **Skin domain complexity.** The 11-class skin classifier achieves the lowest accuracy (~84%), reflecting the fundamental difficulty of fine-grained dermatological classification from photographic images without clinical metadata. Macro F1 (0.775) is notably lower than weighted F1 (0.841), indicating weaker performance on minority classes specifically the inflammatory dermatoses (Atopic Dermatitis, Eczema, Psoriasis) which are visually overlapping and difficult to distinguish even for clinicians without a full history.
- **Caries recall.** The Caries class in the teeth domain has only 36 test samples and achieves a recall of 0.53 despite precision of 1.00 the model misses about half of true Caries cases. This is the single most clinically important failure mode in the pipeline. Mitigation requires sourcing additional Caries training data, ideally with consistent close-up intraoral modality.
- **No external clinical validation.** All reported accuracy figures are sample statistics on held-out splits drawn from the same source datasets used for training. They are not estimates of clinical performance. Real-world deployment would require external validation on independently collected clinical cohorts with verified pathology ground truth, ideally drawn from multiple institutions to test cross-site generalisation.
- **Domain routing artefact sensitivity.** The domain classifier may partially rely on source-dataset artefacts (lighting, camera type, crop patterns, post-processing) rather than purely anatomical features. Routing accuracy on out-of-distribution clinical images (e.g. a phone photograph taken with different lighting) could be lower than the 99.45% test-set figure suggests.
- **Abstention coverage.** 11-17% of predictions are flagged as low-confidence and abstained. In a clinical screening context this is the correct behaviour, but it represents an opportunity cost relative to a less honest system that would surface confident-looking predictions across the board.

- **No multi-task joint training.** The current architecture trains four independent models. A multi-task architecture with shared backbone weights and separate heads might offer better feature transfer and a lower memory footprint, at the cost of more complex training and harder per-domain optimisation. This is an explicit future-work item.

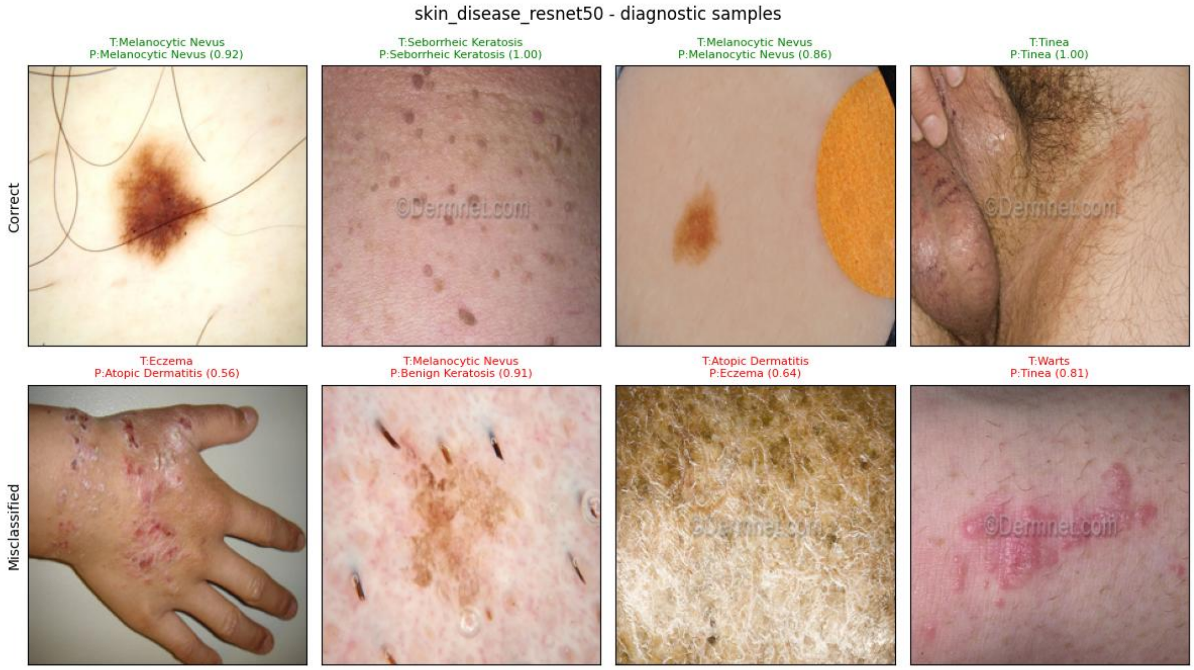

Figure 26. *Diagnostic grid for the ResNet50 skin classifier. Top row: high-confidence correctly classified examples (Melanocytic Nevus, BCC, Tinea, Normal). Bottom row: informative misclassifications illustrating visual overlap between the inflammatory dermatoses.*

4. Discussion and Conclusions

This project successfully demonstrates the feasibility of a unified, multi-domain medical image screening system using a two-stage CNN inference pipeline. The system achieves 85.91% end-to-end accuracy on a held-out test set of 5,648 images spanning 24 disease classes across three medical domains, exceeds 99% on the domain routing task, and is deployed as a responsive, privacy-preserving web application with interpretability features (Grad-CAM, severity triage, plain-language explanations) and safety mechanisms (calibrated abstention, mandatory disclaimers). Four technical findings deserve emphasis.

Architecture comparison validates ResNet50 as production backbone. ResNet50 outperformed VGG16 on every single task, with the largest absolute gain on the hardest task (11-class skin, +1.76 pp) and on the integrated end-to-end pipeline (+1.88 pp). The architectural reasons residual skip connections enabling deeper representation learning, 4x richer feature maps, and better calibration on the confidence-coverage trade-off align with the broader medical-imaging AI literature. Retaining VGG16 only as a benchmark comparison is well supported by the data.

Data quality dominates architecture choice. The dramatic improvement from the Part C baselines (43-79%) to the final production results (83-96%) was driven less by architecture choice than by data-quality work: perceptual-hash deduplication that removed 5,855 duplicate images, class balancing through capping / upsampling / class-weight compensation, focal loss for the imbalanced skin domain, and Neural Style Transfer augmentation for the under-represented eye classes. This is consistent with the modern data-centric AI thesis: in well-explored architectures, marginal returns on data-quality work substantially exceed marginal returns on architecture innovation.

Interpretability is achievable without accuracy cost. Grad-CAM heat-maps consistently localised to clinically meaningful regions (lesion boundaries in skin, iris area in eye, gum line in teeth) across all three disease classifiers, with no observable accuracy degradation. Interpretability features can be added to a production system at essentially zero cost, contradicting the older accuracy versus interpretability trade-off framing. Combined with the severity triage tier and the plain-language explanation, every prediction is paired with a defensible justification a prerequisite for any medical AI tool.

Calibrated abstention is a first-class safety mechanism. The calibrated thresholds lift accuracy on the confident subset by 5-8 percentage points compared with unconditional test-set accuracy. When the system says it is confident, it is meaningfully more likely to be correct than the headline figure suggests. Surfacing the abstention status directly in the GUI (red pill below threshold) translates a statistical property into a concrete user-experience improvement.

Future extensions of the system fall into five categories:

- **External clinical validation.** Partnering with a clinical institution to evaluate MediVision on independently collected cohorts with verified pathology ground truth. This is the single highest-priority follow-up because all reported metrics are currently sample statistics on the source datasets and do not estimate real-world clinical performance.
- **Architecture modernisation.** Exploring more recent architectures (EfficientNet-V2, ConvNeXt, Vision Transformers) for the skin domain where current accuracy is lowest. Vision Transformers have shown strong results on dermatological benchmarks and could plausibly improve skin macro F1 above 0.80.

- **Caries data augmentation.** The single largest clinical limitation - ResNet50s 0.53 recall on the Caries class - should be addressed by acquiring more Caries training data, ideally with consistent close-up intraoral modality.
- **Test-time augmentation in production.** TTA was implemented during evaluation but not yet wired into the production inference path. Averaging predictions across the original image and a horizontal flip can improve test-set accuracy by 0.5-1.5 pp with negligible latency cost.
- **Multi-task joint training.** Re-architecting the system as a single shared-backbone multi-task model with three disease heads could reduce GPU memory footprint by ~70% (one ResNet50 instead of four) and potentially improve feature transfer between domains, though per-domain optimisation flexibility would be reduced.

In summary, MediVision integrates the modern deep-learning workflow from data acquisition and quality filtering through transfer learning, cross-architecture benchmarking, interpretability, and full-stack deployment into a single coherent system that achieves credible accuracy on a non-trivial multi-domain medical imaging task. The pipeline design (two-stage routing with domain-specific classifiers) and the production architecture choice (ResNet50 over VGG16, justified by both accuracy and calibration metrics) are both defensible against the empirical results. The system is not, and should not be, a clinical diagnostic tool but as a research and educational prototype it demonstrates that the technical building blocks for accessible AI-assisted medical screening are mature and ready for the next stage of clinical evaluation.

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ISIC 2019 Skin Lesion Images for Classification

Skin Diseases Image Dataset

Eye Diseases Classification Dataset

Dental Radiography Dataset

X-Ray Dental Cavity Dataset

Oral Diseases Dataset

Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., Ghemawat, S., Goodfellow, I., Harp, A., Irving, G., Isard, M., Jia, Y., Jozefowicz, R., Kaiser, Ł., Kudlur, M., … Zheng, X. (2016). *TensorFlow: Large-scale machine learning on heterogeneous distributed systems*. arXiv. https://arxiv.org/abs/1603.04467

Appendices

A. Individual Contribution

Arnav Mehta (14552020)

Led the end-to-end deep-learning model development for both backbone architectures (ResNet50 and VGG16). Designed and implemented the two-stage CNN inference pipeline including the domain router and three domain-specific disease classifiers, training all eight models on the NVIDIA A100 GPU in Google Colab Pro. Built the checkpoint system with atomic weight writes, architecture-fingerprint validation, and Google Drive mirroring to survive Colab session preemptions. Implemented the rigorous evaluation framework bootstrap 95% confidence intervals, calibrated abstention threshold tuning per domain, top-k accuracy reporting, per-class precision / recall / F1, and one-vs-rest AUC-ROC. Implemented the 16x16 grayscale average-hash perceptual deduplication that removed 5,855 duplicate images, and the VGG19-based Neural Style Transfer augmentation pipeline for the under-represented eye classes. Built the Keras 3-compatible Grad-CAM module that handles the nested ResNet50 submodel structure, and the post-hoc symptom-refinement module with negation handling and capped probability adjustment. Coordinated dataset acquisition from Kaggle (5 sources) and Roboflow (3 sources via REST API), and authored the deep-learning sections of this final report.

Arpita Pandit (24603351)

Designed and implemented the MediVision web application from scratch, including the Landing Page, Image Upload Screen, Results Screen, How It Works section, and About section. Authored the complete frontend stack using vanilla HTML, CSS, and JavaScript with no framework dependency, ensuring zero build-step deployment. Created the clinically appropriate design system (deep teal #1A4A5E and medical green #7ECFB3 palette, Times New Roman headings, amber medical-disclaimer banners) and implemented the responsive layout that adapts to mobile, tablet, and desktop viewports. Replaced pictographic emoji with monochrome SVG line icons in the polish pass for a cleaner clinical aesthetic. Built the drag-and-drop file upload zone with MIME-type and 10 MB size validation, the file preview card with loaded-status indicator, and the symptom textarea. Implemented the Grad-CAM heat-map rendering, the Top Predictions ranked bar chart, the severity triage component, and the first-visit medical disclaimer modal with sessionStorage gating. Authored the GUI Design submission (Part D) and the GUI / Results-presentation sections of this final report.

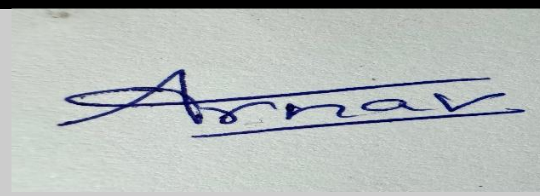
Yash Hingu (14552032)

Managed dataset sourcing, exploratory data analysis, and the Part C initial baseline experiments across all three medical domains. Identified, downloaded, and catalogued the seven public sources (ISIC 2019, HAM10000-derived dermatology datasets, ODIR-5K, Mendeley Eye Diseases Classification, and the dental radiography / oral diseases sets) and authored the unified dataset documentation. Conducted the Part C initial baseline experiments comparing Custom CNN, ResNet50, and InceptionV3 across the three domains, providing the performance floor against which the final ResNet50 and VGG16 results are benchmarked. Performed exploratory data analysis (class distribution visualisation, image resolution profiling, per-domain sample inspection). Authored the Project Registration Proposal (Part A) and the Dataset Details / GUI

Design / Implementation Plan document (Part B). Assisted with final model evaluation (confusion-matrix interpretation, ROC analysis) and final report compilation including the architecture-comparison tables, results discussion, and references.

B. Individual Contribution Split

All team members have discussed and agreed on the above individual contribution to the project.

Team Member	Contribution	Signature
Arnav Mehta	**40%**	
Arpita Pandit	**30%**	
Yash Hingu	**30%**